Two observables of one wall: how surface relaxivity can bias the diffusion intra-axonal fraction and the myelin water fraction

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Abstract

Purpose: Surface relaxivity and time-dependent diffusion are two readouts of the same wall collisions on one substrate. The transverse rate microstructure imaging fits is a bulk rate plus a surface rate $\rho\,(S/V)$; intra- and extra-axonal water carry *different* S/V, so their T_2 differ, biasing any compartment estimate that normalises by a TE-weighted $b = 0$. We quantify the resulting bias on the diffusion intra-axonal signal fraction and on the myelin water fraction (MWF).

Methods: Closed forms for the interior (Brownstein–Tarr) and exterior (Novikov–Burcaw) surface rates over myelinated cylinders are derived and validated by wall-counting Monte Carlo. A calibre-distributed forward model, and grid-free closed forms, give the intra-axonal fraction (f_{intra}) bias vs packing and echo time and the MWF bias vs calibre; a within-subject demyelination trajectory traces the longitudinal case.

Results: The interior/exterior S/V ratio is $(1 - \text{VF})/(g\,\text{VF})$, independent of the calibre distribution, crossing unity at fibre volume fraction $\text{VF}^* = 1/(1 + g)$. Physiological white matter sits above it, so surface relaxivity *over*-weights the intra-axonal signal — the leading-order intra-axonal fraction bias — by ≈12% over the robust packing band at clinical PGSE (TE = 80 ms, cited ρ). That fractions are TE-dependent is known; what is new is the surface-relaxivity and packing *attribution* and the closed-form sign law, with a testable packing-dependent TE drift. The same physics reads through relaxometry as a smaller MWF bias: the thinnest axons' water crosses below the myelin window and is counted as myelin, so fine WM reads myelin-richer (∼0.33 pp at cited ρ, far beneath single-voxel noise and exposed only by regional averaging, but super-linear in the order-of-magnitude-uncertain ρ). In primary (lumen-preserving) demyelination the MWF bias is lumen-set, nearly constant, and cancels in the change.

Conclusion: One S/V surface-relaxivity physics biases both diffusion and relaxometry microstructure estimates; the f_{intra} bias is first-order and packing-dependent (a spatially structured systematic, scaling with the uncertain ρ), the MWF bias small and structural.

Keywords: intra-axonal signal fraction; myelin water fraction; surface relaxivity; transverse relaxation; Monte Carlo simulation; white-matter microstructure

1 Introduction

White-matter microstructure imprints more than one MR contrast, and sometimes the same physics drives two of them. When a water molecule diffuses among myelinated axons it repeatedly strikes their walls, and each collision does two things at once: it interrupts the molecule's displacement — the event whose statistics, accumulated over the fibre packing, make the extra-axonal diffusion coefficient time-dependent[1,2] — and it exposes the spin to the wall's relaxation sink, adding *surface relaxivity*[3] at a rate proportional to the pore surface-to-volume ratio S/V. Surface relaxivity and time-dependent diffusion are, in this sense, two readouts of one wall-collision process on one substrate. A diffusion gradient resolves that substrate's *orientational* structure — the fibre orientation distribution (FOD) — and its *temporal* structure, $D(t)$; the gradient-free multi-echo spin echo reads its scalar, orientation-integrated projection, the rate $\rho\,(S/V)$, fused into what the experiment calls "T_2".

This shared origin touches both communities, and the usual assumption that diffusion is "relaxivity-blind" is only half true. A same-echo-time b=0 normalisation cancels the *common* surface factor $e^{-\mathrm{TE}\,\rho(S/V)}$, so the total signal and the fibre orientation are indeed blind to it — but intra- and extra-axonal water sit against *different* walls with different S/V, and that *differential* T_2 does not cancel. It reweights the compartments in the very b=0 that every diffusion microstructure model normalises by, biasing the recovered **intra-axonal signal fraction** f_{intra} (NODDI, spherical-mean, standard model). The same S/V physics reads through relaxometry as a bias in the **myelin water fraction (MWF)** — the short-T_2 component of the multi-echo (CPMG / GRASE) decay, one of the most widely used myelin indices[4–6], fit with a non-negative least-squares (NNLS) T_2 spectrum and a fixed myelin-water window[6,7]. The transverse rate the MWF is fit to is a *material* bulk term $1/T_{2,\mathrm{bulk}}$ plus the *geometric* surface term $\rho\,(S/V)$ (Eq. (2)); both isotropic and TE-linear, a non-identifiable combination no multi-echo measurement can separate (Sec. 2.2). Surface relaxivity shortens the apparent intra/extra-axonal T_2 toward the myelin window; for the thinnest axons it drags that T_2 *below* the window, where the water is counted as myelin, so fine white matter of *identical myelin volume* reads myelin-*richer*. Both effects are distinct from, and additive to, the known microstructure dependences of these metrics — B_1/flip angle[7,8], iron and susceptibility, and inter-compartment exchange (itself an S/V-driven, axon-size-dependent MWF bias[9,10]); to our knowledge neither has been reported as a surface-relaxivity term inseparable from the bulk T_2.

We develop this on analytical footings and confirm it by simulation. We derive closed forms for the interior (Brownstein–Tarr) and exterior (Novikov–Burcaw) surface rates over the myelinated-cylinder FOD, and validate them with a wall-counting Monte Carlo that contains *no* analytical relaxivity model — a parameter-free boundary-local-time estimator consistent with the established simulator MCMRSimulator[11] (App. A). We then carry the rate to the two microstructure metrics. For the diffusion intra-axonal fraction, a closed-form packing law — the interior/exterior S/V ratio is $(1 - \mathrm{VF})/(g\,\mathrm{VF})$ — over-weights the intra-axonal signal (the leading-order f_{intra} bias) by ≈12% on physiologically dense white matter at clinical echo time, with a testable, packing-dependent TE drift. That compartment fractions drift with TE when their T_2 differ is itself established[12]; our contribution is to attribute that difference to a surface-relaxivity rate set by calibre and packing, and the closed-form sign law that follows. For relaxometry, the same physics gives the smaller MWF bias (fine WM reads myelin-richer, super-linear in the poorly-known ρ). Finally we ask whether the surface term corrupts a longitudinal (lumen-preserving) demyelination measurement, and this is where the g-ratio unifies the two metrics: the MWF bias is lumen-set (the interior wall), hence nearly constant and *cancels* in the change, whereas the f_{intra} bias — set by the interior-*minus*-exterior differential — *drifts* strongly as the exterior wall retreats with the myelin. One S/V physics, longitudinally opposite readouts: the relaxometry change is a cross-region confound only, but a diffusion f_{intra} change across a demyelinating tract carries a large surface artefact.

Throughout we adopt the simplification the standard MWF analysis already makes — instantaneous, ideal refocusing pulses — so that only bulk T_2 and surface relaxivity are in play. In particular we exclude susceptibility, the *other*, genuinely orientation-dependent fibre-angle dependence of MWF[13]; it and finite-pulse coherence effects are orthogonal to the degeneracy shown here and treated elsewhere.

2 Theory

2.1 The myelin water fraction and surface relaxivity

Myelin-water imaging reads the myelin water fraction (MWF) — one of the most widely used non-invasive indices of myelin content — from the multi-echo spin-echo decay, decomposing it with a non-negative T_2 spectrum into a short-T_2 pool (water trapped between the myelin bilayers, $T_2 \approx 10\,\mathrm{ms}$) and a long-$T_2$ intra/extra-axonal (IE) pool ($T_2 \approx 50$–$80\,\mathrm{ms}$), and reporting the short-T_2 signal fraction[4–7].

The decay it fits, however, is not set by the molecular environment alone. A water molecule diffusing among myelinated axons repeatedly strikes their walls, and each collision shortens its transverse coherence — *surface relaxivity*[3] — at a rate proportional to the pore surface-to-volume ratio S/V. This geometric rate $\rho\,(S/V)$ adds to the bulk relaxation the fit is meant to read, shortens the IE pool's apparent T_2 towards the myelin window, and is set by axon calibre and packing — microstructure the relaxometry experiment cannot resolve. Two voxels of *identical myelin volume* but different calibre or packing therefore report different MWF. The rest of this section makes that precise: the surface term is formally inseparable from the bulk T_2 (Sec. 2.2); it is the relaxometric face of the same wall collisions diffusion MRI reads as time-dependent diffusion (Sec. 2.3); and it has closed forms over the myelinated-cylinder distribution (Sec. 2.4).

2.2 The apparent transverse rate is non-identifiable

Two processes shorten the transverse magnetisation between the ideal refocusing pulses. The first is intrinsic (*bulk*) relaxation $1/T_{2,\mathrm{b}}$, set by the molecular environment. The second is the surface loss introduced above — microscopically, paramagnetic surface sites and the field inhomogeneity of the interface: in the fast-diffusion (motional-narrowing) limit, where a spin samples the whole pore many times within an echo period, Brownstein and Tarr showed it is mono-exponential at a rate proportional to the surface-to-volume ratio[3],

$$\frac{1}{T_{2,\mathrm{surf}}} = \rho \frac{S}{V}, \tag{1}$$

with ρ the (transverse) surface relaxivity, a property of the wall.

The two processes add, so the rate the experiment actually decays at is

$$R_2^{\mathrm{app}} = \frac{1}{T_{2,\mathrm{b}}} + \rho \frac{S}{V}. \tag{2}$$

Crucially, the two terms of Eq. (2) are *not separately observable in a multi-echo measurement*, for two simultaneous reasons:

1. **They share the same signal subspace.** Both terms are *isotropic* — surface relaxivity, like bulk relaxation, acts whenever the magnetisation is transverse, with no gradient or orientation dependence — and both accrue *at a constant rate in echo time*: in the motional-narrowing limit the surface attenuation is $\exp(-\rho\,(S/V)\,t)$, the same single-exponential form as $\exp(-t/T_{2,\mathrm{b}})$. For a single pore size they are exactly degenerate; a pore-size *distribution* makes the surface term mildly multi-exponential, but its leading effect is a mean shift $\rho\,\langle S/V\rangle$ that is itself degenerate with the bulk rate, and the residual curvature does not lift the degeneracy in a standard fit.

2. **The geometry term is unobserved.** The surface term depends on S/V, a property of the sub-voxel geometry that a relaxometry experiment does not resolve — many

substrates of different calibre or packing give near-identical decays — so $\rho\,(S/V)$ cannot be predicted and subtracted, and ρ itself is not independently established for myelin. It is an *unobserved* addition to what the fit reports as $1/T_2$.

In short: what the fit calls T_2 is already a mixture of the material $1/T_{2,\mathrm{b}}$ and a geometric $\rho\,(S/V)$ that no multi-echo measurement can disentangle — and this holds even with no diffusion-encoding gradient, since the spins strike walls regardless.

2.3 One wall-collision process, two observables

The surface term in Eq. (2) is not an exotic addition; it is the quantitative-MRI face of a process the diffusion-MRI community already characterises in detail. Novikov, Fieremans, Burcaw and colleagues[1,2,14] showed that in randomly packed media the *extra-axonal* diffusion coefficient is time-dependent: as the diffusion time grows, walkers encounter more of the disordered packing, and the ensemble-averaged $D(t)$ relaxes towards its tortuosity limit D_∞ (the long-time plateau diffusivity) at a rate set by the two-point density correlation function $\Gamma(\mathbf{r})$ — how the cylinder positions correlate across the pack —

$$D(t) \;\approx\; D_\infty + \frac{\Gamma_0}{2\pi}\,\frac{\ln(4D_\infty t/e\zeta^2)}{t} \qquad \text{(2D random cylinder packing)}, \tag{3}$$

with Γ_0 and ζ the disorder strength and correlation length. The diffusion experiment reads this off as a b-value-encoded, time-dependent attenuation.

The same wall collisions that produce the time dependence of $D(t)$ are, from the qMRI side, surface-relaxivity events. *Every* encounter of a spin with a wall both (i) interrupts its free displacement — the event whose statistics, summed over the packing, give $D(t)$ — and (ii) exposes it to the wall's relaxation sink, accumulating surface local time. It is the *same walls*, counted two ways. How tightly the two rates share a functional object depends on the regime. At *short* times both are governed by the small-wavevector limit of the same packing correlation function $\Gamma(\mathbf{r})$: the Mitra $t^{-1/2}$ surface term of the relaxation is the literal twin of the $\sqrt{t}$ correction to $D(t)$[14,15]. At the *long* times a myelin-water train runs, however, $D(t)$ still carries Γ's shape (its approach to the tortuosity plateau, Eq. (3)) whereas the surface rate has saturated to the motional-narrowing plateau $\rho\,\langle S/V\rangle$ — set by the *mean* exterior surface density (the normalisation of Γ), not its shape. So the honest statement is not "one correlation function, two observables" at all times, but: two readouts of one wall-collision process on one substrate, sharing a common short-time origin and a common mean geometry. The diffusion measurement plays a gradient and so resolves that substrate's *orientational* structure (the fibre orientation distribution) and its *temporal* structure $D(t)$; the relaxometry measurement plays no gradient and reads only the scalar, orientation-integrated projection, the rate $\rho\,(S/V)$.

This is literal in the spherical-convolution picture of white matter. The voxel is one fibre orientation distribution (FOD) $\mathcal{F}(\hat{\mathbf{n}})$ convolved with one per-fibre response kernel $\mathcal{K}$[16],

$$S(\hat{\mathbf{g}}, b, \mathrm{TE}) = \int_{S^2} \mathcal{F}(\hat{\mathbf{n}})\,\mathcal{K}(\hat{\mathbf{g}}\cdot\hat{\mathbf{n}}; b, \mathrm{TE})\,\mathrm{d}\hat{\mathbf{n}}, \qquad \mathcal{K} = \sum_{c\in\{\mathrm{i},\mathrm{e},\mathrm{m}\}} f_c\, e^{-\mathrm{TE}\,R_{2,c}}\, E_c(\hat{\mathbf{g}}\cdot\hat{\mathbf{n}}, b), \tag{4}$$

each compartment carrying its transverse rate $R_{2,c}$ — $R_{2,\mathrm{i}} = 1/T_{2,\mathrm{b}} + \rho\,\langle 4/d\rangle_V$ (interior wall), $R_{2,\mathrm{e}} = 1/T_{2,\mathrm{b}} + \rho\, S_{\mathrm{ext}}/V_{\mathrm{ext}}$ (exterior; Sec. 2.4) — and its diffusion attenuation E_c. The surface-relaxivity rate is gradient-free, so it sits entirely in the $\ell = 0$ monopole of this kernel, $\kappa_0(\mathrm{TE}) = \sum_c f_c\, e^{-\mathrm{TE} R_{2,c}}$: a multi-echo experiment carries the surface term in full (fused into the $R_{2,c}$) but is blind to the FOD's shape, which is why relaxometry reports a scalar $\rho\,(S/V)$ with no fibre-angle dependence (absent susceptibility).

Standard diffusion processing, conversely, normalises that factor away. A diffusion experiment divides by its own $b{=}0$ image at the *same* echo time, so the common relaxation factor $e^{-\mathrm{TE}(1/T_{2,\mathrm{b}}+\rho S/V)}$ cancels and the FOD it estimates is, to leading order, *relaxivity-blind* — exact for a single compartment, with a multi-compartment voxel leaving only a second-order

TE-reweighting of the compartment fractions,

$$E(\hat{\mathbf{g}}, b) = \sum_c w_c(\mathrm{TE})\, E_c(\hat{\mathbf{g}}, b), \qquad w_c(\mathrm{TE}) = \frac{f_c\, e^{-\mathrm{TE}\, R_c}}{\sum_{c'} f_{c'}\, e^{-\mathrm{TE}\, R_{c'}}}, \tag{5}$$

the same effect that makes the apparent intra-axonal fraction TE-dependent[12]. The surface term is thus the one substrate property that relaxometry cannot separate from the bulk T_2 (Sec. 2.2) and that diffusion divides out — carried by both signals, resolved by neither, yet set by the same wall geometry. Whether a tailored diffusion acquisition could nonetheless recover that geometry is a separate question we return to, and qualify, in the Discussion.

2.4 Closed forms over the myelinated-cylinder distribution

A myelinated axon is two concentric walls: the spins of the intra-axonal compartment diffuse in the lumen and bounce on the *inner* wall at the inner (axonal) diameter d, while the extra-axonal spins bounce on the *outer* myelin surface at diameter d/g, with g the g-ratio. White-matter inner diameters follow a Gamma distribution $P(d) \propto d^{\alpha-1}e^{-d/\beta}$ with shape $\alpha \approx 2$[17,18]; the outer-diameter distribution is the same scaled by $1/g$. The two walls therefore set two contributions to the surface-relaxivity attenuation, one keyed to d and one to d/g.

Interior wall (intra-axonal water). For a cylinder of inner diameter d, the surface-to-volume ratio is $S/V = 4/d$. Spins are populated by cross-sectional area ($\propto d^2$), so the compartment attenuation is the *area-weighted* ("spin-weighted") average of the Brownstein–Tarr factor over $P(d)$ — the same population weighting the AxCaliber diffusion signal uses[18]. With $S/V = 4/d$ the relevant moment is

$$\left\langle \tfrac{4}{d} \right\rangle_V = \frac{\int 4/d\; d^2 P(d)\, \mathrm{d}d}{\int d^2 P(d)\, \mathrm{d}d} = \frac{4}{\beta(\alpha+1)}, \tag{6}$$

so the area-weighted attenuation is a modified-Bessel form in t with effective rate $\rho\, \langle 4/d \rangle_V$.

Exterior wall (extra-axonal water). Extra-axonal spins bounce on the outer (myelin) surfaces, at diameter d/g, of the surrounding cylinders — so the exterior surface density $S_{\mathrm{ext}}/V_{\mathrm{ext}}$ carries the g-ratio. Its surface-relaxivity rate is governed by the *same* disordered packing that sets the time-dependent diffusion of Eq. (3) — sharing its short-time Γ-controlled behaviour and, as we now show, saturating at long times to a plateau fixed by the mean exterior surface density.

At *short* diffusion times the exterior rate follows the Mitra $t^{-1/2}$ surface-to-volume law[15,19] from the same small-wavevector correlations $\Gamma(k) \sim k^p$ ($p = 0$ for randomly packed cylinders) that set $D(t)$[14] — the short-time relaxometric twin of the Mitra $D(t)$ expansion. Multi-echo myelin-water trains, however, run to *long* echo times ($t \gg \zeta^2/D_\infty$) in the *fast-diffusion* (motional-narrowing) regime[3], where $\rho\,\ell/D_\infty \ll 1$; both hold comfortably for myelinated white matter (with $\zeta \sim 1\,\mu$m and $D_\infty \sim 1\,\mu\mathrm{m}^2$/ms the crossover $\zeta^2/D_\infty \sim 1$ ms sits far below the 10–320 ms echo times, and $\rho\,\ell/D_\infty \approx 10^{-3}$). A walker therefore samples the full exterior geometry many times per echo, the rate saturates to a constant plateau,

$$\frac{1}{T_{2,\mathrm{surf}}^{\mathrm{ext}}} = \rho^{\mathrm{ext}}\,\frac{S_{\mathrm{ext}}}{V_{\mathrm{ext}}}, \tag{7}$$

identical in form to the interior term and to Eq. (1). Two distinct statistical objects are at play here, and it is worth separating them cleanly. The long-time plateau rate is a *one-point* quantity — the *mean* exterior surface density $S_{\mathrm{ext}}/V_{\mathrm{ext}}$ (number density $\times$ per-cylinder outer perimeter over extra-axonal area). It equals $\rho\,\langle S/V\rangle$ because, in the fast-diffusion limit, the lowest Robin eigenvalue of the connected extra-axonal domain reduces to ρ times its total surface over total volume whenever the corresponding eigenmode is uniform across the pore network — i.e. under motional narrowing over the whole extra-axonal space. The *approach* to that plateau, by contrast, is governed by the *two-point* density correlation $\Gamma(\mathbf{r})$ (Eq. (3)) — the same object that sets the time-dependence of $D(t)$, and whose small-wavevector limit gives the short-time Mitra twin. So the two observables share the two-point Γ in their *time-dependence* and share only the one-point mean geometry at the *plateau* the multi-echo train integrates over. The mean-field plateau is moreover exact only when the extra-axonal water motionally averages over the *local* distribution of S/V within an echo; where tight near-contacts kinetically isolate high-S/V pockets the uniform-eigenmode assumption fails and the arithmetic mean is a lower bound (by Jensen, the convex Robin response to a distribution of local S/V exceeds the response to the mean), a limit we return to in Sec. 5. Both walls therefore contribute an isotropic, TE-linear Brownstein–Tarr rate to Eq. (2), tied through d and g to the substrate geometry the relaxometry experiment does not resolve.

For the ratio of the two walls the calibre distribution drops out entirely. Writing the exterior density as total outer-wall perimeter over extra-axonal area, $S_{\text{ext}}/V_{\text{ext}} = \sum 2\pi r_{\text{out}}/(L^2 - \sum \pi r_{\text{out}}^2)$, and using $r_{\text{in}} = g\, r_{\text{out}}$ with the fibre (outer) volume fraction $\text{VF} = \sum \pi r_{\text{out}}^2/L^2$, the interior-to-exterior ratio reduces to a purely geometric factor,

$$\frac{\langle 4/d \rangle_V}{S_{\text{ext}}/V_{\text{ext}}} = \frac{1 - \text{VF}}{g\,\text{VF}}, \tag{8}$$

independent of the calibre distribution, which sets only the common scale. The two rates cross at $\text{VF}^* = 1/(1+g)$: below it the interior wall dominates, above it the extra-axonal water — squeezed into thin, high-S/V crevices between tightly packed fibres — relaxes faster. This single ratio governs the intra-axonal signal-fraction bias of Sec. 4.2.

3 Methods

A wall-counting Monte Carlo shows the multi-echo rate absorbs $\rho\,(S/V)$ with no gradient (Sec. 3.2); that validated rate is then carried, over the full calibre distribution, into two microstructure readouts on a fixed-myelin substrate — the diffusion intra-axonal signal fraction (Sec. 3.3) and the standard myelin-water analysis (Sec. 3.5). All constants are in Table 1.

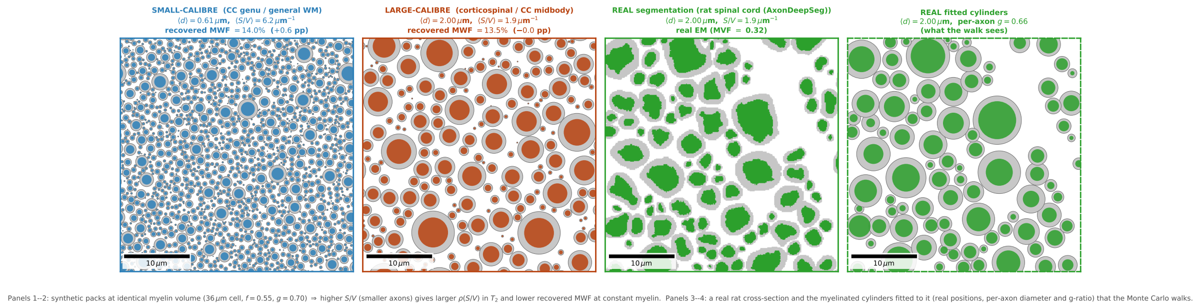

Figure 1: White-matter substrates: synthetic packs at fixed myelin volume, and a real cross-section beside the cylinders fitted to it. *Panels 1–2:* small- and large-calibre random packs of Gamma-distributed myelinated cylinders at the same scale, fibre fraction $f = 0.55$, g-ratio $g = 0.70$, and *the same myelin volume*; small axons pack more wall per unit volume (higher S/V), so the absorbed $\rho\,(S/V)$ is larger and the recovered MWF (annotated, from the Fig. 5 forward model) is lower at identical myelin content. *Panels 3–4:* a real rat spinal-cord cross-section (AxonDeepSeg, $0.13\,\mu$m/px;[20]) and the myelinated cylinders fitted to it (Sec. 3.7; dashed border) — one of the 10 cross-sections whose S/V range is the rug in Fig. 5B.

Reproducible from `figures/fig_substrate_cross_section.py`.

3.1 Canonical white-matter substrate

We act on three substrates (Fig. 1): two synthetic packs that bracket the physiological calibre range and one real rat cross-section. Each synthetic substrate is a random pack of myelinated cylinders with Gamma-distributed inner diameters (shape $\alpha = 2$), g-ratio $g = 0.70$, and fibre (axon+myelin) volume fraction $f = 0.55$, giving a myelin volume fraction $\mathrm{MVF} = f(1 - g^2) = 0.28$ and, with a myelin water content $w_{\mathrm{m}} = 0.40$[21], a true myelin water signal fraction of 13.5%[17,18,22]. *The myelin volume is held fixed throughout.* Axon calibre is swept by changing the Gamma scale (mean inner diameter 0.5–3 μm) at fixed g and f, so the surface-to-volume ratio varies by a factor of six while the myelin volume fraction is identically constant. The relevant interior moment is the *area/spin-weighted* $\langle 4/d \rangle_V = 4/[\beta(\alpha + 1)]$ (Sec. 2.4), not the bare $4/d$. The reference relaxivity $\rho = 1.16\,\mu$m/s is a histology-calibrated model coefficient[23], not an independently measured axolemma relaxivity, so we sweep ρ over 0–2.5 μm/s and report the result as a bound (Sec. 4.3).

Table 1: Constants used. White-matter values are the canonical substrate of the companion modelling work; sources as cited. Orientation dispersion is listed for completeness: being orientation-blind it does *not* affect surface relaxivity or the MWF bias.

Symbol	Quantity	Value
α	Gamma diameter shape	2
d	mean inner diameter (swept)	0.5–3.0 μm
g	g-ratio [24]	0.70
f	fibre volume fraction	0.55
MVF	myelin volume fraction ($= f(1-g^2)$, fixed)	0.28
w_{m}	myelin water content [21]	0.40
ρ	transverse surface relaxivity [23]	1.16 μm/s
$T_{2,\mathrm{b}}^{\mathrm{IE}}$	intrinsic (surface-free) IE bulk T_2	75 ms
T_2^{MW}	myelin water T_2	12 ms
D_0	intrinsic diffusivity	1.7–2.0 μm^2/ms
ODI	orientation dispersion index, typical WM [25,26]	0.1–0.2
CPMG: 32 echoes, TE = 10 ms, ideal 180°		

3.2 Monte Carlo: surface relaxivity by wall counting

The simulator (dmipy-sim) propagates 4×10^4 random walkers through the explicit packed geometry under Brownian dynamics, with specular reflection at the cylinder walls. Surface relaxation is applied inline during propagation: at each wall encounter the walker's transverse magnetisation is multiplied by $\exp(-2\rho\, d_\perp/D)$, where $d_\perp$ is the perpendicular overshoot of the reflected step, and these per-encounter factors accumulate into a walker weight $\exp(-\rho\,\mathcal{L}_{\partial\Omega}/D)$ with $\mathcal{L}_{\partial\Omega}$ the total wall contact — a realised-overshoot estimator of the boundary local time that realises the Brownstein–Tarr Robin condition without any analytical relaxivity model in the propagation (App. A derives the local-time construction, verifies it against the closed form, and confirms it is consistent with the established estimator of[11]). The local time is resolved by sub-stepping the walk to the extra-axonal pore scale (step $\lesssim (S_{\mathrm{ext}}/V)^{-1}/8$): the confined intra-axonal walkers fully sample the inner wall and are accurate at any step, but the fast extra-axonal walkers under-count grazing outer-wall contact at a coarse step, so both compartments converge onto the closed form only when the extra pore is resolved. The Brownstein–Tarr behaviour, if it emerges, emerges from counting wall encounters. The myelin is made impermeable ($\kappa = 0$) so that the intra- and extra-axonal compartments are not contaminated by exchange into the short-T_2 myelin pool, isolating surface relaxivity as the only wall effect. The pure surface factor of each compartment is the mean surviving weight $\langle \exp(-\rho\,\mathcal{L}_{\partial\Omega}/D)\rangle$ over that compartment's walkers (no diffusion gradient, no bulk-T_2 term), and compared against its closed form — the area/spin-weighted Brownstein–Tarr interior term and the Novikov–Burcaw exterior term — over the realised pack, versus echo time at the canonical substrate (Fig. 5A). As a unit check, the same wall-counting on a single cylinder of radius R reproduces the Brownstein–Tarr rate $\rho\,(2/R)$ to $< 1\%$ across radii at the production sub-step.

3.3 Intra-axonal signal-fraction bias

The diffusion intra-axonal fraction f_{intra} is estimated from signal normalised by a b=0 at the same echo time, under the assumption of a shared compartment T_2; surface relaxivity breaks that assumption by giving the interior and exterior walls different S/V (Sec. 2.4), hence different apparent T_2. We compute the resulting bias two ways, which agree to $\lesssim$2 pp. The *exact* calculation carries the full calibre distribution: at diffusion echo times the myelin water has decayed, so the two surviving compartments are the intra-axonal lumen (volume $\propto \mathrm{VF}\,g^2$, relaxing over the distributed interior rate $\rho\,(4/d)$ weighted by water volume $\propto d^2P(d)$) and the extra-axonal water (volume $\propto 1-\mathrm{VF}$, at the exterior rate $\rho\,S_{\text{ext}}/V_{\text{ext}}$); the apparent fraction is the TE-weighted ratio of these attenuated amplitudes. The *closed form* replaces the distributed intra pool by its mean rate, giving the single-exponential odds rescaling of Eq. (9) governed by the calibre-independent ratio Eq. (8). Both are evaluated over fibre volume fraction (0.30–0.85), calibre, and echo time (Fig. 3); no estimator or Monte-Carlo replay is involved, the bias being an analytic consequence of the two validated wall rates.

3.4 Spherical-mean fit and the injected-prior correction

To confirm the signal-weight bias survives an actual estimator (Fig. 4), we forward-simulate a dense white-matter multi-shell diffusion signal ($\mathrm{VF} = 0.68$, 3 shells $b = 1, 2, 3\,\mathrm{ms}/\mu\mathrm{m}^2$, 90 directions each plus 8 b=0, PGSE $\delta/\Delta = 12/40\,\text{ms}$) from the analytical white-matter model with the surface factors on, at echo times 50–90 ms, and fit each with the standard spherical-mean technique (SMT: a stick intra-axonal + zeppelin extra-axonal spherical-mean model[27]) in `dmipy-fit`. The compartment diffusivities are fixed to their known values (leaving the fraction the only free parameter), because a free-diffusivity fit is degenerate on this heavily T_2-reweighted signal. To isolate the *surface* bias from the SMT model-mismatch (a two-compartment stick+zeppelin fit to a three-compartment substrate) we fit the *same* model to a surface-off ($\rho = 0$) forward at each TE; the surface-off recovered fraction is TE-independent (spread <0.01 across 50–90 ms), confirming the TE drift of the surface-on fit is the surface term and not a fitting artefact. The injected-prior correction takes the region's histology calibre prior (fixed α, outer scale, g, f_{axon}, ρ), from which $\rho\,[(S/V)_{\text{int}} - (S/V)_{\text{ext}}]$ is known, and inverts the odds rescaling of Eq. (9) at each single TE.

3.5 Instantaneous-pulse CPMG and the standard MWF estimate

We adopt the standard multi-echo protocol (32 echoes, TE = 10 ms) with ideal, instantaneous 180° refocusing — the conventional assumption, under which the only transverse-decay mechanisms are bulk relaxation and surface relaxivity. The MWF is estimated with the standard chi-square-regularised NNLS T_2 spectrum (zero-order Tikhonov weight set per signal within a factor 1.02 of the unregularised misfit, the Whittall–MacKay / Prasloski / DECAES choice; myelin signal integrated below a 25 ms cutoff)[6,7,28] — the same basic NNLS optimisation the standard pipelines use. The intrinsic IE bulk T_2 is fixed at $T_{2,\text{b}}^{\text{IE}} = 75\,\text{ms}$ (a single material constant), which at the canonical S/V reproduces the literature apparent IE T_2 of $\sim$50–70 ms once the surface term is added.

The decay fed to the estimator is a fixed-myelin forward model carrying the Monte-Carlo–validated rate (Sec. 3.2) as its only surface input; using a forward model rather than raw simulated signal isolates the surface confound from protocol-specific spectral artefacts and, crucially, *retains the full calibre distribution* that the simulator's step-size truncation removes from the raw walk. There is a myelin-water pool ($T_2^{\mathrm{MW}} = 12\,\mathrm{ms}$), an extra-axonal pool at $1/T_{2,\mathrm{b}}^{\mathrm{IE}} + \rho\,(S_{\mathrm{ext}}/V)$, and — the key ingredient — a *calibre-distributed intra-axonal pool*: each inner diameter d in the Gamma population relaxes at $1/T_{2,\mathrm{b}}^{\mathrm{IE}} + \rho\,(4/d)$, water-volume-weighted, so the intra signal is a spectrum rather than a single exponential (a lumped mean rate misses the thin-calibre tail entirely, and reverses the sign of the recovered bias). The amplitudes are fixed by the fixed myelin volume: with fibre fraction $f = 0.55$, $g = 0.70$ and myelin water content $w_{\mathrm{m}} = 0.40$, the water-weighted volumes are myelin $f(1-g^2)w_{\mathrm{m}} = 0.112$, intra (lumen) $fg^2 = 0.270$, and extra $1-f = 0.45$, normalising to the signal fractions $f_{\mathrm{m}} = 0.135$, $f_{\mathrm{ia}} = 0.324$, $f_{\mathrm{ea}} = 0.541$. The bulk T_2 is a fixed material constant, and only the geometric S/V (through calibre) and the unknown ρ vary. We assign intra and extra the *same* intrinsic bulk T_2, so their apparent difference is the surface term alone; a genuine intrinsic intra/extra T_2 difference would add to it, and our bias is the surface contribution on top of any such offset. The robust readout needs no estimator: the short-T_2 signal fraction — myelin water plus the intra water with T_2 below the 25 ms cutoff, i.e. from axons thinner than $d^* = 4\rho/(1/T_2^{\mathrm{cut}} - 1/T_{2,\mathrm{b}}^{\mathrm{IE}})$ — is a grid-free property of the decay, which we report alongside the standard NNLS (which reproduces its sign and amplifies it). Robustness is checked under the chi-square-adaptive weight, fixed zero-order weights (0.01, 0.02), and Rician noise (single-voxel SNR = 150).

3.6 End-to-end check: MWF from the full Monte-Carlo signal

As an end-to-end check, free of any forward model, we simulate the full multi-echo CPMG decay with the forward Monte Carlo (instantaneous 180°; a per-echo gradient-free walk carrying per-compartment bulk T_2 and first-principles wall relaxivity, no analytical decay) and run the same NNLS estimator on it. The intrinsic bulk T_2 is a fixed material value (75 ms) at the actual walls, so turning surface relaxivity on adds $\rho\,(S/V)$ and produces the calibre-dependent apparent decay. Over the long CPMG train the surface local time is sub-stepped at the extra-axonal pore/2 (Sec. 3.2), a coarser resolution than the Panel-A validation but ample for the decay. We walk four calibres (surface on and off on each), add Rician noise at SNR = 200 (150 realisations), and NNLS-fit each. This end-to-end walk reproduces the decay and the IE-pool spectral shift, but its *integrated* short-T_2 fraction barely moves: we deliberately do not walk the sub-micron crossing calibres here (below $\sim 0.4\,\mu$m an impermeable-wall walk needs prohibitively small steps and compute), so the quantitative MWF bias is carried by the full-distribution forward model above (Sec. 3.5).

3.7 Real white-matter cross-sections

To anchor the substrate in real tissue and to test that the operating point was not cherry-picked, we use every manually-segmented cross-section in the AxonDeepSeg SEM dataset (rat spinal cord, 8 animals, 10 cross-sections, 0.07–0.18 μm/px;[20]). Each cross-section is represented as a pack of myelinated cylinders fitted to its segmentation: every segmented axon becomes a cylinder at its true centroid, with inner radius the area-equivalent radius of the lumen and outer radius the area-equivalent radius of the lumen plus its watershed-assigned myelin ring, giving a *per-axon* diameter and g-ratio. This preserves the geometry the physics depends on — the axon-size distribution and packing, hence S/V — while routing the walk through the validated exact-circle reflection (the irregular-segmentation walk does not confine the extra-axonal walkers; the fitted-cylinder representation does). The fitted reconstruction is shown beside the raw segmentation in Fig. 1 for transparency. The interior area-weighted moment $\langle 4/d \rangle_V$ across the 10 cross-sections spans 0.9–2.3 $\mu\mathrm{m}^{-1}$ (the rug in Fig. 5B), confirming that real white matter populates the swept S/V range.

3.8 Demyelination trajectory

To ask how the surface-relaxivity bias behaves as a tract demyelinates, we trace a *within-subject* series rather than compare independently sampled substrates. For each of $N_{\mathrm{subj}} = 24$ "subjects" we draw a Gamma axon population ($\alpha = 2$, canonical calibre, $N = 300$ cylinders) and pack the outer (myelin) cylinders *once*, at the baseline g-ratio $g_0 = 0.70$ — the thickest myelin, hence the largest outer walls; packing the largest-ever cylinders first guarantees every later, thinner state is overlap-free. We then *freeze the centres and the inner (axon) radii* and demyelinate by raising g toward 1 at fixed axon, so the outer wall shrinks ($r_{\text{out}} = r_{\text{in}}/g$) while the axon count and lumen are preserved. Each subject is thus its own longitudinal series on one fixed axon distribution, not a fresh random substrate per stage. At every stage we read the geometry directly off the realised pack — the myelin volume fraction $\text{MVF} = \sum \pi(r_{\text{out}}^2 - r_{\text{in}}^2)/L^2$ (hence the true MWF), the interior $\langle 4/d \rangle_V$ (unchanged, the axon lumen being preserved), and the exterior $S_{\text{ext}}/V_{\text{ext}} = \sum 2\pi r_{\text{out}}/(L^2 - \sum \pi r_{\text{out}}^2)$ (which falls as the outer walls retreat) — and feed the *frozen inner-radius distribution* to the same calibre-distributed forward model as Sec. 3.5, read as the grid-free short-T_2 fraction, at $\rho = 0$ (true MWF) and at the cited ρ (apparent MWF), averaging over subjects. Because the crossing that sets the bias depends on the inner radii — which this trajectory holds fixed — the offset is nearly constant and cancels in the within-subject change; the exterior S/V that halves plays no role in it. No new wall rate is introduced: the surface rate is the one the Monte Carlo validated in Sec. 3.2, applied to the demyelinating geometry. The 24 packs differ only in random cylinder placement, so the spread across them measures packing-realisation variance (negligible here) rather than biological variability; the trajectory is, in effect, deterministic.

4 Results

4.1 Surface relaxivity enters the signal and the apparent T_2

Figure 2A shows the surface attenuation of the intra- and extra-axonal water at $b = 0$ — with no diffusion-encoding gradient — as a function of echo time, on the canonical substrate. The simulator contains no analytical relaxivity model: it propagates walkers, reflects them specularly at the walls, and accumulates the wall-contact local time, so the surface attenuation $\langle \exp(-\rho\, \mathcal{L}_{\partial\Omega}/D) \rangle$ read off the recorded contact is the simulator's first-principles wall dephasing. It lands on the closed-form surface factor for both walls — the interior Brownstein–Tarr term over the realised inner-diameter distribution and the exterior Novikov–Burcaw term (Fig. 2A), removing 36.4% (Monte Carlo) versus 36.6% (closed form) of the intra-axonal amplitude and 34.0% versus 34.0% of the extra-axonal amplitude by TE $= 60\,\mathrm{ms}$ — both walls reproduced to $\lesssim 0.2\,\mathrm{pp}$. (The boundary-local-time estimator is resolved by sub-stepping the walk to the extra-axonal pore scale, so it is converged for both compartments; Sec. 3.2.) This pack-level agreement is an internal consistency check — Monte Carlo and closed form both evaluate the motional-narrowing rate over the same realised geometry, so it confirms bookkeeping rather than the physics. The genuinely independent validation is against the *exact* lowest Robin eigenvalue on a single cylinder (the root of $(kR)J_1/J_0 = \rho R/D$, not merely its leading $\rho\,(2/R)$ limit; App. A): the wall-counting estimator lands on it to $< 1\%$ across radii, down to the sub-micron calibres that matter below — so the rate the next section carries across the calibre distribution is validated at the small-diameter end, not extrapolated. The central physical fact follows from first principles: every wall encounter the Novikov–Burcaw theory counts toward the time-dependent diffusion $D(t)$ is also a surface-relaxation event, and the resulting rate $\rho\,(S/V) \approx 5\,\mathrm{s}^{-1}$ at the operating point enters the transverse decay even though no gradient is present to encode the displacement.

The same physics shows directly in the multi-echo signal. Walking the *full* CPMG train with the forward Monte Carlo — per-compartment bulk relaxation plus first-principles wall relaxivity, with and without surface relaxivity, holding the intrinsic bulk T_2 and the myelin volume fixed (Sec. 3.6) — the surface term visibly speeds the decay (Fig. 2B), and more for the fine, high-S/V substrate than for the coarse one, even though both share the same bulk T_2. Fed to the standard NNLS T_2-spectrum analysis, the apparent long-T_2 intra/extra (IE) pool shifts toward the short-T_2 myelin window (Fig. 2C): surface relaxivity shortens the apparent IE T_2, the more so at high S/V. In the raw walk the *integrated* short-T_2 fraction barely moves — the sub-micron axons whose water actually crosses the window lie below the simulator's calibre floor — so the quantitative myelin-water bias is carried by the full-distribution forward model of the next section, of which this is the estimator-free, spectral-shift counterpart. The

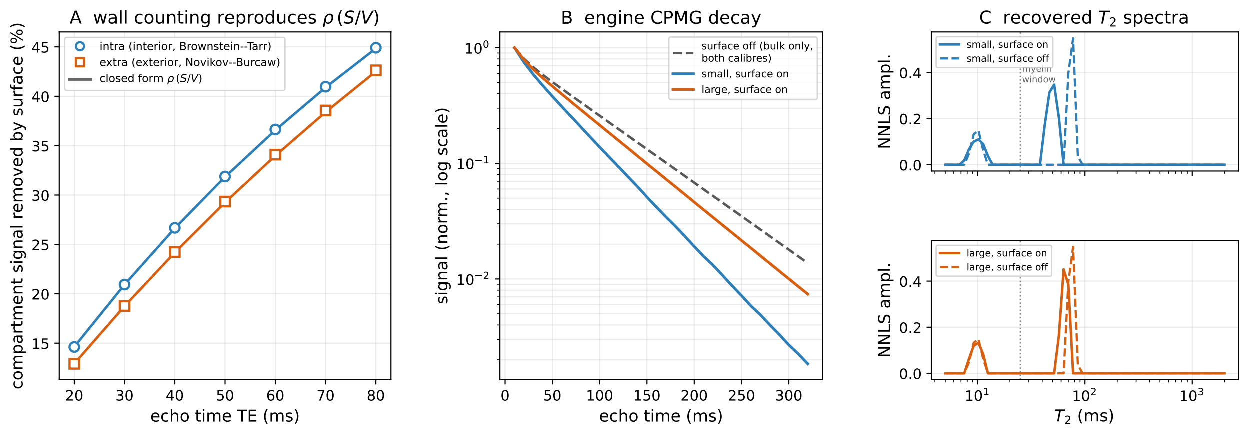

Figure 2: Surface relaxivity enters the multi-echo signal and the apparent T_2. (A) Forward Monte Carlo surface attenuation at $b = 0$ (wall counting, no analytical relaxivity model; markers) versus the closed-form interior (Brownstein–Tarr, circles) and exterior (Novikov–Burcaw, squares) factors (solid lines) — both walls reproduced to $\lesssim$0.2 pp (the engine sub-steps the boundary local time to the extra-axonal pore scale). So wall-counting reproduces $\rho\,(S/V)$, and the multi-echo T_2 carries it with no gradient. **(B)** Engine-simulated gradient-free CPMG decay (log scale), surface on (solid) for small/large calibre vs the calibre-independent surface-off reference (dashed grey): the surface term speeds the decay, more at high S/V. **(C)** Standard-NNLS T_2 spectra, surface on vs off: surface relaxivity shifts the long-T_2 IE pool toward the 25 ms myelin window, more at high S/V. *Reproducible from* `figures/fig_cpmg_mwf_engine.py` (panel A reads `fig_mwf_sv_bias.py`).

apparent T_2 of the intra/extra water is thus $1/(1/T_{2,\mathrm{b}} + \rho\,S/V)$, with the geometric term inseparable from the material one (Sec. 2.2); because $S/V = 4/d$ for the interior wall and scales likewise for the exterior, the shift is set by axon calibre and packing — the dependence the next section carries into the myelin water fraction.

4.2 Surface relaxivity biases the intra-axonal signal fraction

The most consequential reading of that T_2 shift is on the diffusion side. Every diffusion microstructure model — NODDI, the spherical-mean technique, the standard model — estimates an intra-axonal signal fraction f_{intra} and normalises the diffusion-weighted signal by a b=0 acquired at the *same* echo time, assuming the compartments share one T_2. They do not: the interior and exterior walls carry different S/V, so surface relaxivity gives the intra- and extra-axonal water different apparent T_2. To leading order the fraction an unbiased estimator returns is the TE-weighted signal fraction $w_c(\mathrm{TE}) = f_c e^{-\mathrm{TE}R_c} / \sum_{c'} f_{c'} e^{-\mathrm{TE}R_{c'}}$ (Eq. (5)), so the recovered f_{intra} inherits the surface reweighting to leading order. That fractions are TE-dependent when compartment T_2 differ is established[12]; what is new here is the *source* we assign to the compartmental T_2 difference — a surface-relaxivity rate $\rho\,(S/V)$ tied to axon calibre and packing — and the resulting closed-form *sign law*, below. We quantify the effect as the bias in the signal weight w_{intra}; this is the leading-order term that propagates to any fraction estimator. This signal weight is distinct from the *geometric* fibre volume fraction VF that sets S/V: it is a b=0-normalised, T_2-weighted water fraction, related to VF only through the compartment geometry and relaxation, and it is exactly the quantity surface relaxivity biases (which is why the correction injects VF as a prior rather than reading it off the fit). It is not a strict upper bound on the recovered-parameter bias: a fit with the diffusivities constrained returns the weight essentially directly, and estimator model-mismatch can even *amplify* it — our spherical-mean fit below recovers a bias of the same order as, and at points exceeding, the signal-weight estimate. We therefore treat w_{intra} as the representative first-order magnitude, and confirm it with an actual estimator (Fig. 4).

The interior surface-to-volume is the per-axon $4/d$ (area/spin-weighted $\langle 4/d\rangle_V = 2\langle r\rangle/\langle r^2\rangle$, set by calibre alone); the exterior is set by packing — total outer-wall perimeter over extra-axonal area. Their ratio (Eq. (8)) reduces to the purely geometric $(1-\text{VF})/(g\,\text{VF})$, *independent of the calibre distribution* (which sets only the absolute scale), and the two are equal at a crossover packing $\text{VF}^* = 1/(1+g) \approx 0.59$ (Fig. 3A): below it the interior wall dominates and intra water relaxes faster; above it the extra-axonal water, squeezed into thin high-S/V crevices between tightly packed fibres, relaxes faster.

Carried into the fraction estimate, the intra/extra *odds* are rescaled by a single exponential,

$$\frac{f_{\text{intra}}^{\text{app}}}{1-f_{\text{intra}}^{\text{app}}} = \frac{f_{\text{intra}}}{1-f_{\text{intra}}}\,\exp\!\big[-\text{TE}\,\rho\,\big((S/V)_{\text{int}}-(S/V)_{\text{ext}}\big)\big], \tag{9}$$

so f_{intra} is *under*-estimated in loosely packed white matter and *over*-estimated in densely packed white matter, crossing zero at VF^* (Fig. 3B; the closed form of Eq. (9) reproduces the sign law and crossover exactly and tracks the magnitude of the exact distributed-calibre calculation to $\lesssim 4$ pp, slightly *under*-estimating it — by Jensen, the distributed thin-calibre tail leaves the surviving intra water higher than the mean-rate approximation). Physiological white matter (fibre VF ≈ 0.65–0.8) sits *above* the crossover, where at clinical PGSE (TE $= 80$ ms) and the cited ρ the intra-axonal signal weight is over-estimated by $\approx 12\%$ at VF $= 0.65$, rising to $\approx 22\%$ by VF $= 0.72$. At the denser end the exterior S/V approaches $\sim 11\,\mu\text{m}^{-1}$ (sub-0.1 μm crevices) and the mean-field plateau of Eq. (8) becomes marginal (extra-axonal water may not motionally average over the local S/V; Sec. 5), so the largest values are an upper edge, not a central estimate — and, since the local-S/V distribution can only raise the exterior rate (Jensen), the robust $\approx 12\%$ at VF 0.65 is a *lower* bound on the dense side. For traceability: at VF $= 0.65$ the interior and exterior densities are $\langle 4/d \rangle_V \approx 6.3$ and $S_{\text{ext}}/V_{\text{ext}} \approx 8.3\,\mu\text{m}^{-1}$, a differential surface rate $\rho\,(S_{\text{ext}}/V - \langle 4/d \rangle_V) \approx 2.3\,\text{s}^{-1}$ that over TE $= 80$ ms shifts the intra/extra odds by $\approx 20\%$ — a $\approx +9\%$ mean-rate fraction bias, which the exact calibre-distributed calculation (Fig. 3B) raises to $+12\%$ (the $\lesssim 3$ pp closed-vs-exact gap, from the thin-calibre tail). All f_{intra} percentages here are *relative* fraction biases.

The exponent of Eq. (9) is *linear* in ρ, so the same order-of-magnitude uncertainty that bounds the MWF result below applies here directly and symmetrically: the $\approx 12\%$ is quoted at the cited ρ and scales roughly linearly with it (the log-odds shift doubles as ρ doubles). Unlike the MWF term below — which we show rides $\sim 50\times$ beneath *single-voxel* noise — this is a first-order bias on a headline metric; being deterministic in the substrate it is a *spatially structured systematic*, not zero-mean noise, so it does not average down as $1/\sqrt{N}$ but shifts a regional or cohort mean by close to its full value. Whether it clears single-voxel diffusion-fit scatter depends on the protocol and is not established here; its signature is regional and cross-cohort, not per-voxel.

The bias carries a *testable* signature: it grows monotonically with echo time (Fig. 3C), so an f_{intra} that drifts with TE at fixed b is consistent with it. This is necessary but not sufficient evidence — the same TE drift arises from any compartmental T_2 difference (intrinsic

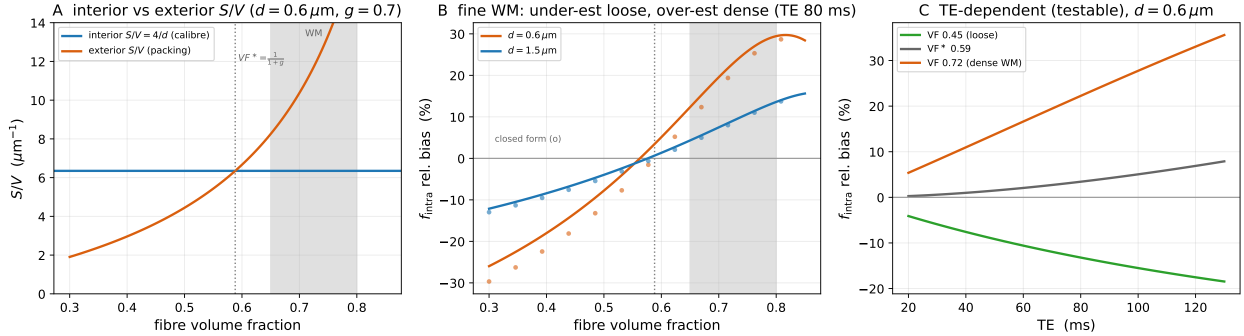

Figure 3: Surface relaxivity biases the intra-axonal signal fraction, with a sign set by packing. (A) Interior S/V (calibre-set, flat) vs exterior S/V (packing-set, rising) vs fibre volume fraction; they cross at $\mathrm{VF}^* = 1/(1+g)$, and physiological WM (grey band) sits above it where the exterior wall dominates. **(B)** Intra-axonal signal-weight w_{intra} relative bias (the leading-order recovered-f_{intra} bias) vs fibre VF at clinical TE $= 80\,\mathrm{ms}$ (lines: exact distributed calculation; circles: closed form Eq. (9)): under-estimated when loose, over-estimated when dense, zero at VF^*; $+12\%$ (relative) at $\mathrm{VF} = 0.65$, rising to $+22\%$ at 0.72. **(C)** The bias grows with echo time — the testable fingerprint — for loose, crossover, and dense packing.

Reproducible from `figures/fig_fintra_bias.py`.

intra/extra bulk T_2, CSF partial volume, exchange), the generic TEdDI effect[12]. What would distinguish the surface term is its *packing dependence*: the sign and magnitude track fibre volume fraction through Eq. (8), reversing below VF^*, whereas an intrinsic-T_2 offset does not.

So far the bias is the reweighting of the intra-axonal *signal weight*; we confirm it survives an actual estimator and that a prior corrects it (Fig. 4). We forward-simulate a dense-WM multi-shell signal with surface relaxivity on, and fit it with the standard spherical-mean technique (SMT: intra stick + extra zeppelin)[27]. Because the surface factor is b-independent it propagates through the whole multi-shell fit, so the recovered f_{intra} inherits the bias and *drifts with echo time* — from $\approx$+15% at TE $= 50\,\mathrm{ms}$ to $\approx$+30% at $90\,\mathrm{ms}$ (the surface term isolated by differencing against a surface-off fit, whose recovered fraction is TE-independent to <0.01, so the drift is the surface term and not a fitting artefact; Sec. 3.4). This confirms the effect is not merely a b=0 prefactor: it propagates to what a real estimator returns, at a magnitude comparable to (here somewhat above) the analytical signal-weight bias. A cohort of healthy controls scanned at different echo times — routine across sites — would thus report spuriously different f_{intra} for identical tissue. The degeneracy that forbids *estimating*

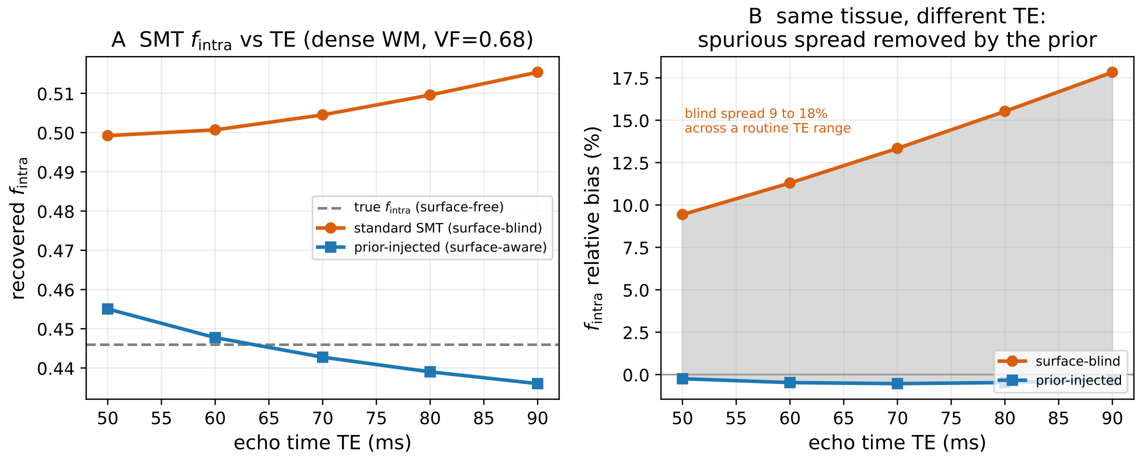

Figure 4: A standard SMT fit returns the surface-biased f_{intra}, and a histology prior corrects it at a single TE. Dense WM (VF = 0.68), analytical WM forward with surface relaxivity on, fit by the standard spherical-mean technique in `dmipy-fit`. **(A)** Recovered f_{intra} vs echo time: surface-blind SMT (orange) drifts well above the surface-free truth (dashed); injecting the histology prior (blue) recovers it. **(B)** The same tissue read at different TE gives a spurious f_{intra} spread of ≈15–30% under a blind fit (a multi-site confound), removed by the prior.
Reproducible from `figures/fig_smt_fintra.py`.

the surface term at one TE (Sec. 2.2) does not forbid *correcting* for it: injecting a region's histology-derived calibre prior (so that $\rho\,[(S/V)_{\text{int}} - (S/V)_{\text{ext}}]$ is known, as NODDI already fixes diffusivities) undoes the odds rescaling of Eq. (9) at a single TE, collapsing the 15–30% drift to a ∼6% residual (Fig. 4) and harmonising the cohort.

4.3 Surface relaxivity also biases the recovered myelin water fraction

The same S/V difference reads through relaxometry too, in the myelin water fraction — a smaller, structural companion that ties the diffusion and relaxometry pictures to one physics. Carrying that surface rate into the standard myelin-water analysis at *fixed myelin volume* (Sec. 3.5), the signal-weighted IE apparent T_2 swings from 68.2 ms for large-calibre (3 μm outer) fibres to 44.9 ms for fine (0.45 μm outer) fibres — a ∼34% change driven entirely by the surface term $\rho\,(S/V)$, with no change in intrinsic tissue T_2 and none in myelin content. This T_2 shift is NNLS-independent (Eq. (2) read directly) and is the paper's most robust quantitative statement.

That shift has a specific, and at first counter-intuitive, consequence for the recovered MWF. The intra-axonal water is not one T_2 but a distribution over calibre: each axon of *inner* (axon) diameter $d_\mathrm{i} = g\,d$ carries an interior $S/V = 4/d_\mathrm{i}$, so its water relaxes at $1/T_{2,\mathrm{b}} + \rho\,(4/d_\mathrm{i})$, and the *thinnest* axons are shortened most. Once the inner diameter falls below a critical calibre $d^* = 4\rho\,/\,(1/T_2^{\mathrm{cut}} - 1/T_{2,\mathrm{b}})$ — 0.17 μm (an inner/axon diameter) at the cited ρ — that axon's water crosses *below* the 25 ms myelin window and is counted, by the window's own definition, as myelin water (Fig. 5A). Fine white matter therefore reads myelin-*richer*, not poorer: the short-T_2 signal fraction, computed grid-free with no NNLS, rises above the true myelin fraction of 13.49% (recovered exactly at large calibre) to ≈14.3% at the finest calibre swept (Fig. 5B); between the reference calibres of the bound below the fine (0.6 μm outer) vs large (2.0 μm outer) excess is ≈0.33 pp at the cited ρ. The standard chi-square-regularised NNLS estimator reproduces the same sign and amplifies it (13.44 → 14.35%; dashed), as it also assigns near-window intra water to the short-T_2 pool. The crossing is not merely a forward-model construction. Walking *isolated* confined axons across a range of inner calibres with the wall-counting Monte Carlo — which is exact for the confined interior (its single-cylinder rate matches the closed form $\rho\,(4/d)$ to the digit and the exact Robin eigenvalue to $< 1\%$, App. A) — recovers an apparent intra-axonal $T_2 = 1/(1/T_{2,\mathrm{b}} + \rho\,4/d_\mathrm{i})$ that falls from 27.4 ms at $d_\mathrm{i} = 0.20\,\mu$m to 22.6 ms at 0.15 μm and 19.2 ms at 0.12 μm — crossing *below* the 25 ms window at $d^* \approx 0.17\,\mu$m (Fig. 6). What the from-scratch simulator cannot yet integrate is a *packed* substrate: a dense pack's calibre floor (step cost $\propto 1/R^2$) truncates the sub-micron tail, so the packed CPMG shows the spectral shift but a negligible integrated MWF drift. The crossing is an interior effect, however, so the single-axon walk establishes it, and the distributed forward model then carries it across the calibre distribution.

Read as myelin *content* — MWF's purpose — the effect is larger than the water-fraction bias. The contaminating signal is intra-axonal water (≈100% water) misassigned to the myelin pool, whose water content is only $W_M{\approx}0.4$ (≈60% dry lipid and protein)[21]; converting apparent MWF to myelin volume at that ratio over-states myelin by $1/W_M{\approx}2.5\times$ the water-fraction bias. The 0.33 pp fine-vs-large MWF excess at the cited ρ thus implies a ≈0.82 pp over-estimate of myelin content between the two regions at equal true myelin (Fig. 5B, right axis).

The magnitude is contingent on ρ, which for myelinated tissue is not independently established — the cited $\rho = 1.16\,\mu$m/s is a coefficient of a relaxation–diffusion model fitted to histology[23], not a measured axolemma relaxivity, and is plausibly uncertain by an order of magnitude. Because d^* grows linearly with ρ while the volume-weighted fraction of axons below it grows super-linearly, the effect climbs steeply: the fine-vs-large MWF excess rises from 0.33 pp at the cited ρ to 3.36 pp at 2.5 μm/s — equivalently, $0.82 \to 8.4$ pp of inferred myelin content (Fig. 5C), no longer negligible at the upper end and a fraction of the 2–5 pp MWF change attributed to demyelination[29–31]. The structural claim (S/V and ρ unobserved, the term inseparable) does not depend on the value. Being deterministic in the substrate the bias does not average away, but at clinical SNR it rides far beneath the *single-voxel* noise: at SNR = 150 a single substrate's recovered MWF scatters by ∼4.7 pp, ∼14× the cited-ρ systematic. That factor compares a single voxel's absolute-MWF scatter against the cross-region excess and so overstates how buried the *contrast* is — some of the NNLS ill-conditioning and realisation variance cancels in a matched fine-vs-large difference. The systematic is nonetheless exposed only by averaging: since it is spatially structured (it tracks calibre and packing) rather than zero-mean, it survives averaging that noise does not, but exposing a ∼0.33 pp offset above a ∼4.7 pp single-voxel floor still needs a region of order $(4.7/0.33)^2 \sim 200$ voxels even in the favourable uncorrelated-noise case. It is thus a small, cohort- or ROI-level systematic, not a per-voxel readout.

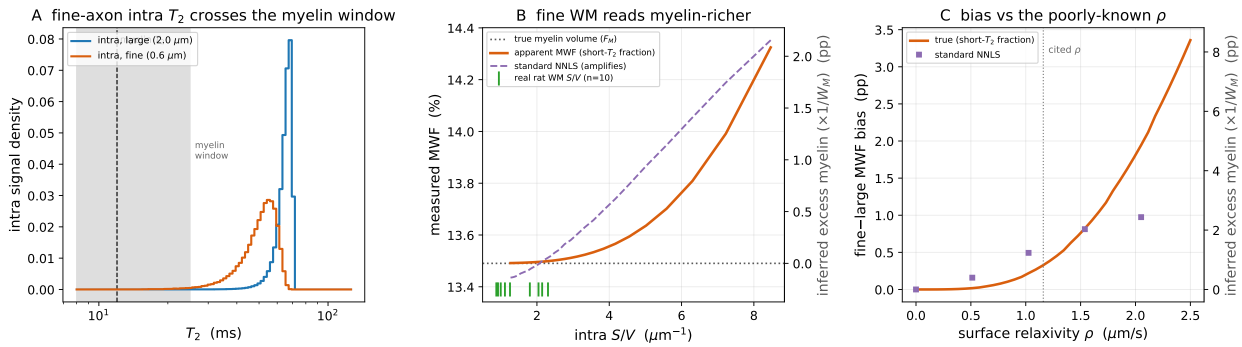

Figure 5: Surface relaxivity biases the recovered myelin water fraction: fine white matter reads myelin-richer. **(A)** Mechanism. Each axon relaxes at $1/T_{2,\mathrm{b}} + \rho\,(4/d)$, so the intra-axonal T_2 is a distribution over calibre; for fine axons ($0.6\,\mu$m, orange) it shifts down and grows a tail that crosses *below* the 25 ms myelin window (shaded), where it is counted as myelin water — the large-calibre pool (blue) does not. **(B)** At fixed myelin volume, the apparent MWF (short-T_2 fraction, grid-free) rises above the true myelin fraction F_M (dotted) as S/V increases — fine WM reads myelin-richer; standard NNLS (dashed) amplifies it. Right axis: inferred excess myelin content ($\times 1/W_M$), the 2.5×-amplified reading. Green rug: interior S/V of 10 real rat cross-sections (AxonDeepSeg; Sec. 3.7). **(C)** The bound: fine- ($0.6\,\mu$m outer) vs large-calibre ($2.0\,\mu$m outer) MWF excess vs the poorly-known ρ — 0.33 pp (≈0.82 pp myelin content) at the cited ρ, climbing super-linearly to 3.4 pp (≈8.4 pp myelin) at $2.5\,\mu$m/s.

Reproducible from `figures/fig_mwf_sv_bias.py`.

4.4 The bias as a tract demyelinates

The bias so far is a cross-region contrast at fixed myelin. The clinically central question is longitudinal: as a tract demyelinates, does the surface term make the apparent MWF *change* mis-state the true myelin loss? We trace this within-subject (24 realised packs; axons frozen, myelin thinned from $g_0 = 0.70$ toward $g = 0.90$; Sec. 3.8), one subject's cross-section at three stages shown in Fig. 7A — the axon centres and inner (lumen) radii frozen, only the outer myelin wall retreating.

The key is that the crossing which causes the bias is set by the *inner* diameter, and primary demyelination preserves the lumen. The bias is therefore a nearly *constant* offset: apparent MWF sits just above true throughout, by +0.34 pp at baseline and +0.30 pp at $g = 0.90$ (Fig. 7B) — the small drift coming only from the water-fraction normalisation as 40%-water myelin converts to 100%-water extra-axonal space, not from the crossing itself. Because a within-subject *change* subtracts the baseline, this near-constant offset cancels:

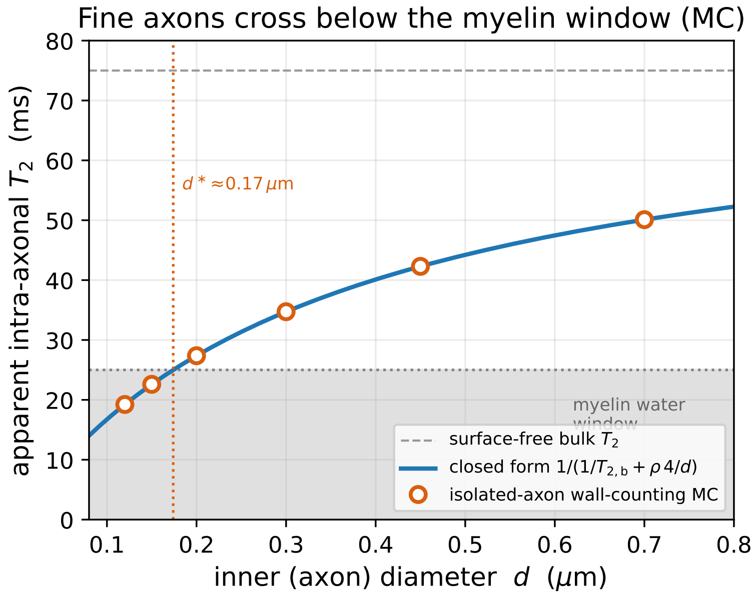

Figure 6: Fine axons cross below the myelin window — shown directly in the Monte Carlo. Gradient-free wall-counting walks in *isolated* reflecting cylinders (confined interior, where the estimator is exact) recover the apparent intra-axonal T_2 (circles) at a range of inner calibres. They sit on the closed form $1/(1/T_{2,\mathrm{b}} + \rho\,4/d)$ (blue) to the digit and cross below the 25 ms myelin window (shaded) at $d^* \approx 0.17\,\mu$m, confirming the mechanism the distributed forward model integrates over the calibre distribution. The packed CPMG cannot reach this sub-micron tail (step cost $\propto 1/R^2$); the interior effect needs only the single-axon walk.

Reproducible from `figures/fig_crossing_mc.py`.

over a trajectory in which the true MWF falls $13.49 \to 2.63\%$ (a 10.86 pp drop) the apparent MWF falls 10.91 pp, an over-report of $\sim$0.05 pp — of order 0.4% of the true change, and within the forward model's own idealisation (Fig. 7C). So the surface-relaxivity confound is a *cross-region* effect, not a longitudinal one: for primary demyelination (a g-ratio rise at preserved axons) it very nearly cancels in same-substrate tracking, because it is tied to the axon lumen the pathology leaves intact. Axonal loss or a shift in the fine-calibre population would break this cancellation — they alter the very inner-diameter distribution the offset is set by — so the statement is bounded to lumen-preserving demyelination.

Crucially, the *diffusion* intra-axonal fraction behaves oppositely along the same trajectory, and the contrast unifies the two halves of this paper through the g-ratio. The MWF bias is set by the *interior* crossing (the inner axon wall, which demyelination preserves), so it is constant and cancels. The f_{intra} bias of Eq. (9) is instead set by the interior-*minus*-exterior differential

$\rho[(S/V)_{\text{int}} - (S/V)_{\text{ext}}]$, and the g-ratio enters both walls: the exterior wall is the fibre boundary at d_{i}/g, so as myelin thins it *retreats* and $(S/V)_{\text{ext}}$ collapses (here $5.4 \to 2.9\,\mu\text{m}^{-1}$ over $g\,0.70 \to 0.90$) while $(S/V)_{\text{int}}$ stays fixed. The differential therefore *grows*, and the f_{intra} bias *drifts* rather than cancels — from $\approx -5\%$ at baseline to $\approx -21\%$ at $g = 0.90$ (Fig. 7D), at a clinical TE = 80 ms. (This trajectory's baseline sits at fibre VF = 0.55, just *below* the crossover $\text{VF}^*(g_0) = 0.59$, so its f_{intra} bias is *negative* — interior-dominated, opposite in sign to the dense-WM headline of Sec. 4.2 — and grows more negative as the exterior wall retreats.) So one S/V physics gives longitudinally opposite readouts: the relaxometry MWF change is nearly confound-free, whereas a diffusion f_{intra} change across a demyelinating tract carries a large, TE-dependent surface artefact — and, because the trajectory sweeps fibre VF across the crossover $\text{VF}^* = 1/(1+g)$, the sign of the f_{intra} bias can even reverse mid-trajectory for a denser healthy baseline.

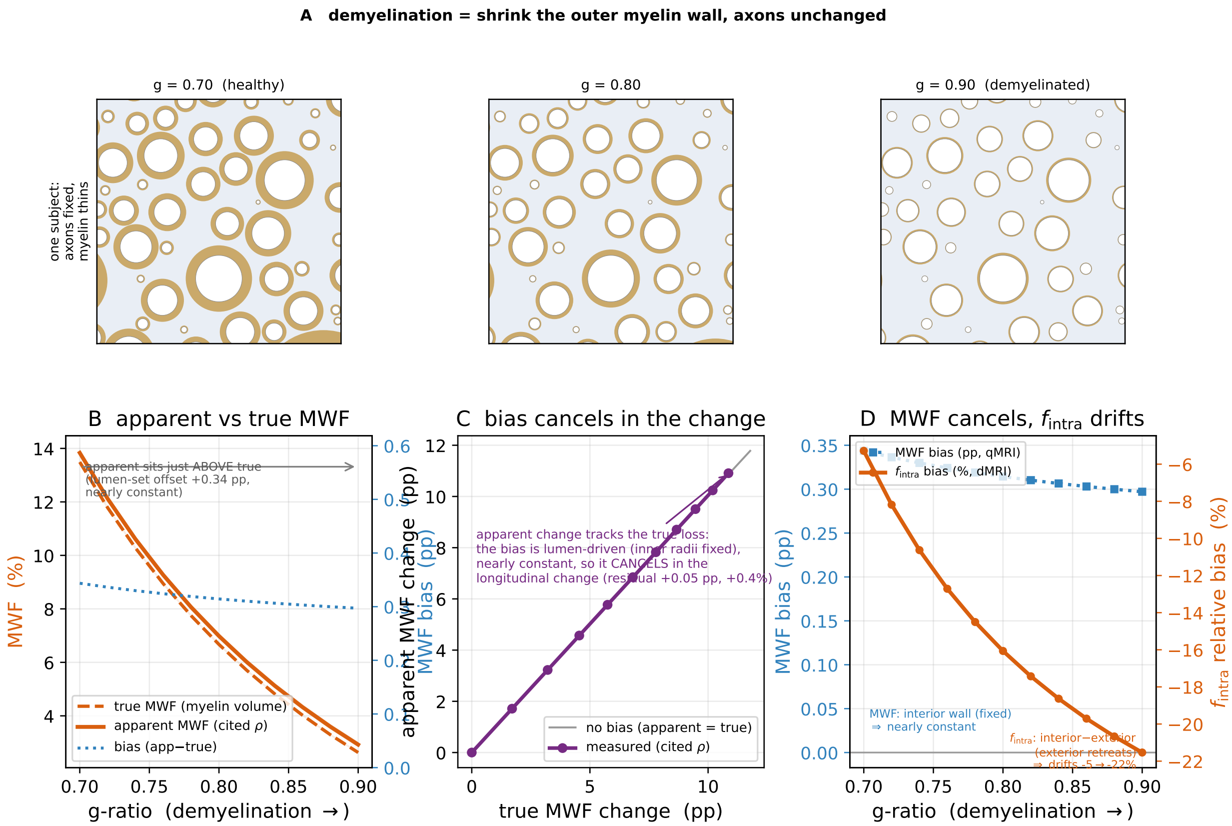

Figure 7: As a tract demyelinates, the surface-relaxivity bias cancels in the longitudinal change. Within-subject trajectory (24 packs; axons frozen, myelin thinned from $g_0 = 0.70$ to $g = 0.90$; Sec. 3.8), same forward model and MC-validated rate as Fig. 5. **(A)** One subject's substrate at three stages: axon centres and inner radii frozen, the outer myelin wall $r_{out} = r_{in}/g$ shrinking. **(B)** Apparent (solid, cited ρ) sits just above true (dashed) MWF, and both fall steeply; the bias (dotted, right axis) is small and nearly constant because it is set by the fixed inner radii, not the retreating outer wall. **(C)** Apparent vs true MWF *change*: the measured curve lies on the no-bias diagonal — the lumen-driven offset cancels, so the apparent loss tracks the true loss to within ∼0.4%. **(D)** The unifying contrast: along the same trajectory the relaxometry MWF bias (blue, left axis) stays flat and cancels — it reads the fixed *interior* wall — while the diffusion f_{intra} bias (orange, right axis) *drifts* from ≈−5% to ≈−21%, because it reads the interior−exterior differential and the exterior wall retreats as myelin thins.

Reproducible from `figures/fig_demyelination_bias.py`.

5 Discussion

The transverse rate a multi-echo spin echo measures is bulk relaxation plus an isotropic, TE-linear surface rate $\rho\,(S/V)$ inseparable from it, with S/V unobserved. The consequence for the myelin water fraction is specific: surface relaxivity shortens the intra/extra apparent T_2, and for the thinnest axons it drags that T_2 below the myelin window, where the water is counted as myelin — so fine white matter reads myelin-*richer* at fixed myelin content, by a substrate-dependent amount ($\sim$0.1 percentage points at the cited ρ, super-linearly more — $> 1\,\mathrm{pp}$ — at the upper end of its order-of-magnitude uncertainty). It is no artefact: a wall-counting Monte Carlo reproduces the rate from first principles, and the short-T_2 crossing is a property of the window's own definition, present before any estimator sees the signal.

The same physics from two communities. The contribution is partly one of perspective: the time-dependent extra-axonal $D(t)$ of randomly packed media is well established in diffusion MRI[1,2,32], and the same wall collisions, seen from relaxometry, are surface-relaxation events (Sec. 2.3). The corollary: the "T_2" a relaxometry experiment reports is contaminated by a geometry-dependent surface rate with no gradient played.

Implications for myelin water imaging. MWF is read as a myelin-content index and compared across tracts, subjects, and time. Because per-voxel MWF is noisy, studies average it over a region of interest — a callosal subregion, a tract, lobar white matter — and compare ROI means; subtracting two such means is where the surface term bites. Each ROI carries its own offset set by its own thin-axon population, so

$$\Delta\mathrm{MWF}_{\mathrm{app}} = \underbrace{\Delta\mathrm{MWF}_{\mathrm{true}}}_{\text{myelin difference}} + \big[\,\beta_A - \beta_B\,\big], \qquad \beta \;=\; F_I\,\Phi(\rho), \tag{10}$$

where F_I is the intra-axonal signal fraction and $\Phi(\rho)$ the volume-weighted fraction of that intra-axonal water dragged below the myelin window (the thin-calibre tail), so β is larger for the finer region. The second bracket — a pure calibre difference — adds to the myelin difference and cannot be separated from it within the multi-echo data. It vanishes only when the two ROIs share their fine-axon population, and is largest where calibre differs systematically: between callosal subregions (the finer genu and splenium versus the large-diameter mid-body[17] — predicting the finer genu/splenium read myelin-*richer* than the mid-body at equal myelin), across development or ageing, or between patient and control groups when disease shifts the calibre spectrum. At $\sim$0.1 pp ($\sim$0.24 pp of inferred myelin content) at the cited ρ it is a small fraction of a gross regional contrast, but a meaningful one for the subtle, $\sim$1 pp differences cross-sectional myelin studies pursue — and, being deterministic in the substrate, a cohort-level systematic that shifts a group mean rather than averaging out. The comparison it spares is the field's cleanest, the *same region of the same subject over time*: there the offset is a nearly-constant, lumen-set term that cancels in the change (Sec. 4.4), because primary demyelination leaves the inner diameters the crossing depends on intact. None of this invalidates MWF; it bounds a contribution of the same S/V-driven family as the documented exchange bias[9], not previously quantified as a surface-relaxivity term inseparable from the bulk T_2.

What counts as myelin water. The mechanism is not really about axons: the short-T_2 window captures *any* water whose T_2 is dragged below it by surface relaxivity, which happens once S/V exceeds $(1/T_2^{\text{cut}} - 1/T_{2,\text{b}})/\rho \approx 23\,\mu\text{m}^{-1}$ at the cited ρ (a sub-micron scale) and $\approx 11\,\mu\text{m}^{-1}$ at the upper ρ. This is the very physics that makes bilayer-trapped myelin water short-T_2 in the first place, so MWF has always measured "high-S/V trapped water", of which myelin is merely the dominant source in health. It follows — mechanism-predicted, not simulated here — that *any* membrane-dense sub-micron compartment qualifies: myelin debris, fine reactive-glial processes or microvacuolation could add short-T_2 signal counted as myelin and prop up apparent MWF, while whole cells ($S/V < 1\,\mu\text{m}^{-1}$) are far too large to cross. This widens the caveat from a calibre confound to a statement about what the short-T_2

pool contains, and is a natural target for a drop-in Monte Carlo of explicit pathological microstructure.

Idealised packing, and where Monte Carlo becomes essential. Our closed forms are exact for the substrate we assume — parallel, circular, non-touching cylinders packed by random sequential adsorption — and for that geometry they are also the *efficient* route. A Monte-Carlo pack pays a sub-step cost that scales as $1/R^2$ in its smallest axon (an impermeable-wall relaxivity weight needs steps $\ll R$), so the very calibres that dominate the bias are the most expensive to simulate[33], while the analytics returns them for free. The idealisation has a specific blind spot, though. The exterior rate uses the *mean* S/V — total wall perimeter over total extra-axonal area — and a mean cannot see the *distribution of local* S/V that realistic packing creates: where axons deform, touch, undulate or cross[34], extra-axonal water is trapped in thin interstitial crevices of very high local S/V, and the window-crossing the intra-axonal calibre tail undergoes could occur there too — extra-axonal water read as myelin. Whether it does hinges on motional averaging, and two *distinct* length scales must not be conflated. A local sheet of high S/V shortens the T_2 enough to cross the myelin window once its width falls below $\sim$90 nm ($S/V \gtrsim 23\,\mu\mathrm{m}^{-1}$ at the cited ρ) — a scale the extra-axonal space genuinely reaches. But whether that water *averages* with the bulk is a *separate* condition, set by how fast a spin escapes the pocket: at a 90 nm width the traversal time $w^2/D \sim 4 \times 10^{-3}$ ms is far below the echo spacing, so an *open* 90 nm sheet averages comfortably and feels only the mean S/V. Averaging fails only where the pocket is kinetically *isolated* — narrow throats or near-contacts whose bottleneck traversal time approaches the echo spacing — a percolation-scale property of the packing, not of the sheet width, and one we do not quantify here. So the mean-field rate is a lower bound (Jensen; Sec. 2.4) and the size of the correction is set by this unquantified isolation scale. This is the regime the analytics cannot reach and only a Monte Carlo of non-idealised geometry can resolve — and, by the same $1/R^2$ argument, the most costly to run. The reading is therefore a duality, not a ranking: the closed form should carry the idealised packing, where Monte Carlo is wasteful, and Monte Carlo should be reserved for the irreducibly non-idealised part — crevices, undulation, crossings — where the closed form has nothing to say.

Why the correction injects the fraction rather than fitting it. The quantity a diffusion model returns is the intra-axonal *signal* fraction, not the geometric fibre volume fraction VF; a spherical-mean fit even ties its extra-axonal (tortuosity) diffusivity to that signal fraction, so the packing geometry that sets S/V is never an independent output. Because the surface term is b-independent and inseparable from the bulk T_2, S/V — and hence VF — is not identifiable from the fit, and feeding the fitted fraction back into the surface term would be circular (it is the very quantity the surface term biases). The correction therefore treats VF as a *bounded prior* supplied from regional histology — exactly the ingredient a cross-TE cohort harmonisation needs and can plausibly obtain — and leaves the fit to return only the corrected fraction.

What could resolve it, and what cannot. The degeneracy is structural, so no refinement of the multi-echo acquisition alone breaks it: echo spacing, train length and signal-to-noise change precision, not the non-identifiability. Diffusion is, in principle, the orthogonal axis — the same wall collisions, read with a gradient. The elegant point is that the disorder enters diffusion through the *same* fibre orientation distribution $\mathcal{F}(\hat{\mathbf{n}})$ as the signal convolution Eq. (4): in the low-b, short-time regime the apparent diffusivity along $\hat{\mathbf{g}}$ is that FOD convolved with a disorder kernel,

$$\frac{D(t;\hat{\mathbf{g}})}{D_0} = \int_{S^2} \mathcal{F}(\hat{\mathbf{n}})\, \kappa_{\mathrm{dis}}(\hat{\mathbf{g}}\cdot\hat{\mathbf{n}};t)\, \mathrm{d}\hat{\mathbf{n}}, \quad \kappa_{\mathrm{dis}} = 1 - \tfrac{4}{3\sqrt{\pi}}\sqrt{D_0 t}\left[\textstyle\sum_c f_c\,(S/V)_c\right]\left[1-(\hat{\mathbf{g}}\cdot\hat{\mathbf{n}})^2\right], \tag{11}$$

with $(S/V)_c$ the interior $\langle 4/d\rangle_V$ or exterior $S_{\mathrm{ext}}/V_{\mathrm{ext}}$. Because $1-(\hat{\mathbf{g}}\cdot\hat{\mathbf{n}})^2 = \frac{2}{3}[1-P_2(\hat{\mathbf{g}}\cdot\hat{\mathbf{n}})]$ carries only $\ell = 0$ and $\ell = 2$, the surface disorder is a *rank-2* function on the sphere — round (ℓ=0) to relaxation, fibre-shaped (ℓ=0+2) to diffusion — and spherical convolution preserves that band limit for *any* FOD. The orientation-invariant total S/V is then the $\ell = 0$ (trace) part, which an isotropic (trace) oscillating-gradient acquisition or a spectrally tuned b-tensor[35,36] recovers independent of fibre dispersion. In practice it is gradient-limited: the short-time regime needs OGSE frequencies whose deliverable b-value collapses as $\sim 1/f^3$ [37] — sitting at the corner frequency of a micron axon needs $G \sim 70$–$260\,\mathrm{T/m}$, one to two orders beyond even preclinical inserts — so the fine calibres that carry the effect lie out of reach, and a realisable protocol is left to future work. What no route removes is the absolute calibration: the inner- and outer-wall relaxivities $\rho_\mathrm{i}, \rho_\mathrm{e}$ — which may differ, and scale the bias — are not independently established for myelinated tissue, and the myelin-water pool itself, with its very large bilayer S/V, is plausibly surface-dominated in a way we have not modelled here.

Relation to exchange and other MWF confounds. MWF is already known to depend on factors other than myelin content: B_1/flip-angle imperfections, which the extended-phase-graph fit corrects[7,8]; iron and susceptibility; and inter-compartment water *exchange*, which Harkins, Dula and Does showed makes the apparent MWF depend on axon size[9,10]. The exchange bias is, like ours, driven by surface-to-volume ratio, and both are largest at high S/V. They are *mechanistically distinct* — exchange moves magnetisation between pools of different T_2, whereas surface relaxivity destroys coherence at the wall and, for the thinnest axons, moves intra water across the window — but *operationally entangled*: both act at the small-calibre end and neither is separable within the multi-echo data. We deliberately isolate surface relaxivity by making the myelin impermeable in the simulation, so the rate validated in Fig. 2A and the bias of Fig. 5 are exchange-free; we do not claim to have measured their *relative* magnitude, which would require permeable walls and is left to future work. Myelin is a strong barrier (slow apparent exchange, 1–$10\,\mathrm{s}^{-1}$, permeation at nodes and paranodes[38]), whereas wall *encounters* recur on the perpendicular diffusion correlation time (~ 0.1–$1\,\mathrm{ms}$): a spin bounces two to three orders of magnitude more often than it permeates, which is what makes isolating surface relaxivity with $\kappa = 0$ physically justified.

Biological idealisations. Four substrate assumptions bound the biological reading. First, the effect is carried by the sub-micron calibre tail (the crossing is at $d^* \approx 0.17\,\mu$m at the cited ρ), so the magnitude depends on how much sub-micron water the tissue holds — which we anchor with rat spinal-cord cross-sections[20], a coarser-calibre tissue than the human cortical and callosal white matter that is the clinical target. Human callosal fibres run finer, with a substantial sub-micron population[17], so if anything the rat-cord anchor *under*-states the tissue-borne effect; a human-EM calibre census is the proper anchor and we do not claim one. Second, we hold the g-ratio fixed at 0.70 across calibre, whereas fine axons are relatively more thickly myelinated (lower g)[24]; since the crossover $\mathrm{VF}^* = 1/(1+g)$ depends on g, the single sign law is a calibre-averaged statement, and a realistic $g(d)$ would spread the crossover across calibres rather than abolish it. Third, we assign the inner and outer walls one relaxivity $\rho_{\mathrm{i}} = \rho_{\mathrm{e}}$; these are chemically different membranes whose relaxivities may differ, a *structural* (not merely scale) uncertainty that could shift the interior/exterior balance and hence the f_{intra} sign, and ρ itself is imported from a histology-fit model coefficient[23] that may partly absorb exchange or susceptibility. Fourth, the longitudinal cancellation assumes lumen-preserving (pure g-ratio) demyelination; real MS and Wallerian degeneration co-vary axon calibre through swelling, beading, atrophy and outright axonal loss, which alter the very inner-diameter distribution the offset is set by — so the clean within-subject cancellation is an idealised bound, and where pathology shifts the calibre spectrum the cross-region confound re-enters the longitudinal comparison.

Limitations. We assumed ideal, instantaneous refocusing, isolating the bulk-versus-surface question; finite-pulse and susceptibility effects are real but orthogonal and treated elsewhere. This sets the effect in perspective: the flip-angle/stimulated-echo and fibre-orientation/susceptibility[13] systematics the MWF field already corrects for are themselves of order a percentage point or more — comparable to or larger than the surface bias at the cited ρ, which is a further, hitherto-unnamed term of the same family. The MWF magnitudes come from a forward model carrying the validated rate over a calibre-distributed intra pool (Sec. 3.5), so the absolute numbers inherit its two-/three-pool, non-exchanging, canonical-Gamma idealisations — which modulate the magnitude but not the existence or sign of the term. The robust core needs no estimator: the closed-form rate and its wall-counting Monte-Carlo validation (a motional-narrowing limit and a geometry-only simulation with no relaxivity model) agree, and the short-T_2 crossing they imply is a grid-free property of the signal. The claim we stand behind is the inseparability itself, the sign of its consequence (fine reads myelin-richer), and the order of magnitude that follows once ρ is fixed.

6 Conclusion

The transverse rate underlying microstructure imaging is an intrinsic bulk rate plus a surface-relaxivity rate $\rho\,(S/V)$ — both isotropic and TE-linear, hence non-identifiable, with S/V unresolved. It is the relaxometric face of the wall collisions diffusion MRI reads as time-dependent extra-axonal diffusion, and it acts with no gradient; a wall-counting Monte Carlo reproduces it from first principles. Because the interior and exterior walls carry different S/V, the compartments relax differently, and any estimate that normalises by a TE-weighted b=0 is biased. On the diffusion side this over-weights the intra-axonal signal (the leading-order intra-axonal fraction bias) on physiologically dense white matter — by ≈12% over the robust packing band at clinical echo time and the cited ρ, with a sign set by a single packing ratio $(1-\mathrm{VF})/(g\,\mathrm{VF})$ that crosses unity at $\mathrm{VF}^* = 1/(1+g)$, and a testable, packing-dependent TE drift — a first-order, systematic bias on a headline metric that shifts regional and cohort means rather than averaging out. The same physics reads through relaxometry as a smaller, structural myelin-water bias: the thinnest axons' water is dragged below the myelin window and counted as myelin, so fine white matter reads myelin-*richer* at equal myelin content (∼0.33 pp, ∼0.82 pp of inferred myelin after the $1/W_M$ amplification, climbing super-linearly at the upper end of the order-of-magnitude-uncertain ρ); being lumen-set, it *cancels* in the within-subject longitudinal change that primary demyelination traces. Diffusion is in principle the orthogonal axis that could constrain the geometry directly — the same wall collisions, read with a gradient — but realising it is gradient-limited, left to future work.

A Surface relaxation as a boundary-local-time estimator

The wall-counting of Sec. 3.2 is a Monte-Carlo realisation of the Brownstein–Tarr Robin boundary condition $D\,\partial_n M = -\rho\, M$ [3]. That boundary condition is not a per-collision penalty: it is a continuous absorbing wall, and its stochastic representation is the *boundary local time* $\mathcal{L}_{\partial\Omega}$ of reflected Brownian motion — the (measure-theoretic) contact the walker accumulates at the surface — through the Feynman–Kac survival weight $\exp(-\frac{\rho}{D}\,\mathcal{L}_{\partial\Omega})$ [39]. Any faithful simulator therefore only has to *estimate* $\mathcal{L}_{\partial\Omega}$ from the discrete walk; different estimators agree in the $\Delta t \to 0$ limit but differ at finite step.

We use the *realised-overshoot* estimator: a step that crosses the wall contributes $2\,d_\perp$ to $\mathcal{L}_{\partial\Omega}$, with $d_\perp$ the perpendicular overshoot of that step, so the transverse magnetisation is scaled by $\exp(-2\rho\, d_\perp/D)$ per encounter — a *pathwise* estimator weighted by the actual penetration depth. An alternative, used by the multi-modal simulator MCMRSimulator (v1.0.0) of Cottaar et al. [11], assigns every detected collision the *same* cost $\exp(-x\sqrt{\Delta t})$ (`src/evolve.jl`, lines 566–570), i.e. it replaces the per-step penetration with its mean; its documented material rate is $x = \rho\sqrt{\pi/D}$, which we use as given (no calibration). Both are legitimate; they differ only in whether the collision cost carries the realised depth or a fixed $\sqrt{\Delta t}$.

To isolate the *estimator* — rather than two codebases, which would confound the per-collision rule with stepping, detection and RNG — we run one reflecting walk in a cylinder cross-section ($R = 3\,\mu$m, $D = 3\,\mu\text{m}^2$/ms) and accumulate *both* penalties on the *same* collisions, referenced to the exact lowest Robin eigenvalue (the root of $(kR)J_1(kR)/J_0(kR) = \rho R/D$, of which the textbook $\rho\,(2/R)$ is only the $\rho R/D \to 0$ limit, larger by $\rho R/4D$). This is an *estimator* comparison inside one code, not a two-codebase cross-validation: reimplementing the fixed-$\sqrt{\Delta t}$ rule on our own walk holds the stepping, detection and RNG common, so the only independent reference is the exact Robin eigenvalue — and it is against that eigenvalue, not against either simulator, that correctness is judged. Both schemes reproduce it at a fine step (Fig. 8B,C), confirming our surface term is implemented correctly; the $\sqrt{\Delta t}$ rule being MCMRSimulator's documented choice, this also shows the two are consistent in the fine-step limit, but it is not an agreement between two independent codes. We adopt the realised-overshoot form (the factor 2 is the local-time normalisation, nothing fitted) because it is parameter-free and holds at coarser steps, where the fixed-$\sqrt{\Delta t}$ rule drifts to -8%.

The estimator's relative error is a function of the two dimensionless groups $\rho R/D$ and step/R alone, so the accuracy shown at $R = 3\,\mu$m transfers to any calibre: holding both groups fixed and sweeping the radius from 3 down to $0.1\,\mu$m leaves the error flat at -0.01% (a scale-invariance check in the same script). This is what licenses carrying the interior rate to the sub-micron calibres that dominate the bias — they are not an extrapolation but the *same* dimensionless regime, and in fact an easier one: at the physical ρ a sub-micron axon has $\rho R/D \sim 10^{-5}$, far deeper in fast diffusion than the benchmark, where the estimator (and the Brownstein–Tarr rate itself) is most accurate (panel C).

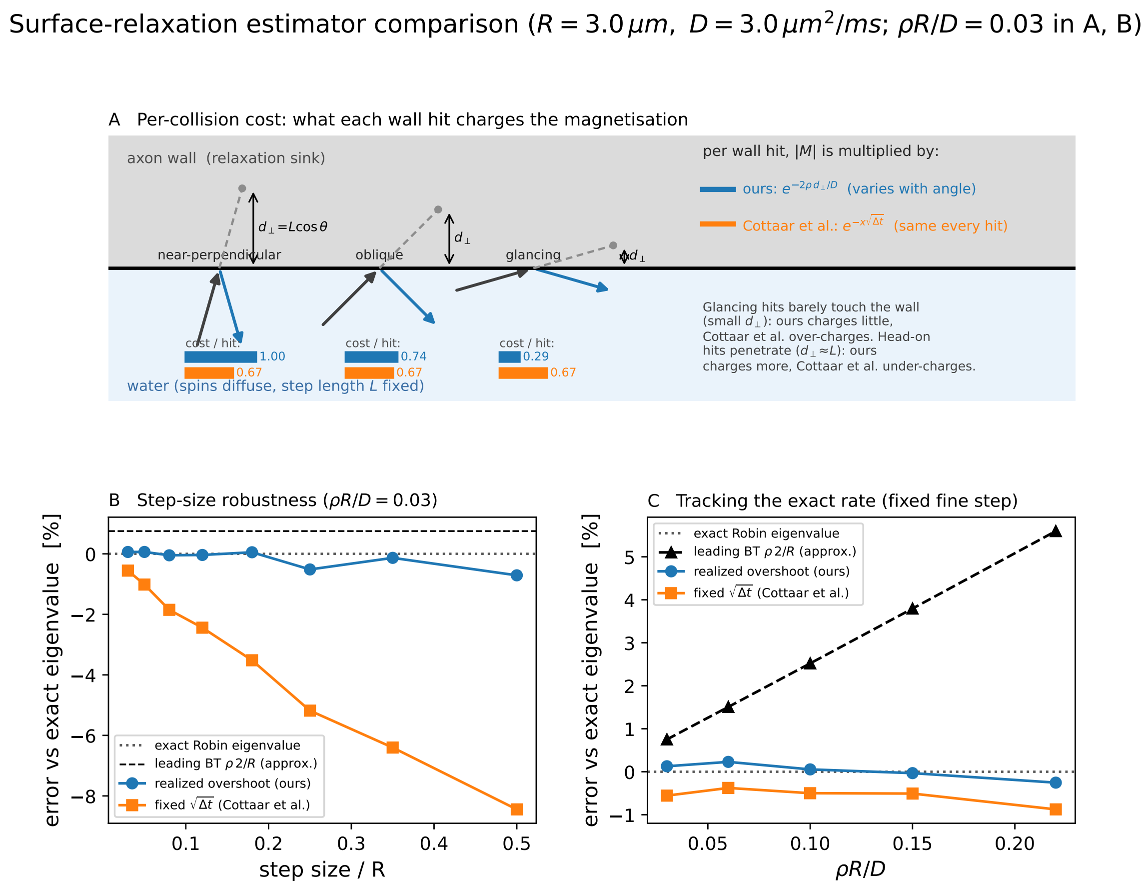

Figure 8: Two Monte-Carlo estimators of the same Brownstein–Tarr surface relaxation, on identical collision events of one reflecting walk. **(A)** Per wall hit: for a fixed step L the perpendicular overshoot $d_\perp = L\cos\theta$ grows from glancing to head-on, so the realised-overshoot rule charges $e^{-2\rho d_\perp/D}$ (blue, 1.00/0.74/0.29) while the fixed-$\sqrt{\Delta t}$ rule charges the same each hit (orange, 0.67) — its mean over-charges glancing and under-charges head-on hits. **(B,C)** Error vs the exact Robin eigenvalue (dotted; leading $\rho(2/R)$ dashed): the parameter-free realised-overshoot estimator (ours) stays on the exact value across step size (B) and $\rho R/D$ (C); the fixed-$\sqrt{\Delta t}$ rule ([11]) drifts to -8% at a coarse step (B); and the leading-order *analytic* formula — not the estimators — diverges to $+5.6\%$ as $\rho R/D$ grows (C).

Data and Code Availability

In keeping with the openness aims of the `dmrai-lab` project, all materials — the manuscript source, the figures, and the Python scripts that regenerate every figure from scratch (each with its committed numeric data) — are openly available at `github.com/dmrai-lab/dmipy`. The figures are built on the dmipy computational ecosystem: `dmipy-fit`, a subsumption and expansion of the original `dmipy` toolbox[40] (the analytical multi-compartment signal models and the NNLS myelin-water estimator), and `dmipy-sim`, a GPU/JAX-accelerated toolbox for physics-complete Monte-Carlo spin simulation under arbitrary free waveforms, sharing one pulse-sequence and substrate interface with `dmipy-fit`. The real cross-sections are the AxonDeepSeg SEM rat dataset[20].

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References

[1] D.S. Novikov, J.H. Jensen, J.A. Helpern, and E. Fieremans. Revealing mesoscopic structural universality with diffusion. *Proceedings of the National Academy of Sciences*, 111(14):5088–5093, 2014. doi: 10.1073/pnas.1316944111.

[2] L.M. Burcaw, E. Fieremans, and D.S. Novikov. Mesoscopic structure of neuronal tracts from time-dependent diffusion. *NeuroImage*, 114:18–37, 2015. doi: 10.1016/j.neuroimage.2015.03.061.

[3] K.R. Brownstein and C.E. Tarr. Importance of classical diffusion in NMR studies of

water in biological cells. *Physical Review A*, 19:2446–2453, 1979. doi: 10.1103/PhysRevA.19.2446.

[4] A. MacKay, K. Whittal, J. Adler, D. Li, D. Paty, and D. Graeb. In vivo visualization of myelin water in brain by magnetic resonance. *Magnetic Resonance in Medicine*, 31: 673–677, 1994. doi: 10.1002/mrm.1910310614.

[5] C. Laule, I.M. Vavasour, S.H. Kolind, D.K.B. Li, T.L. Traboulsee, G.R.W. Moore, and A.L. MacKay. Magnetic resonance imaging of myelin. *Neurotherapeutics*, 4:460–484, 2007. doi: 10.1016/j.nurt.2007.05.004.

[6] K.P. Whittall, A.L. MacKay, D.A. Graeb, R.A. Nugent, D.K.B. Li, and D.W. Paty. In vivo measurement of T2 distributions and water contents in normal human brain. *Magnetic Resonance in Medicine*, 37(1):34–43, 1997. doi: 10.1002/mrm.1910370107.

[7] Thomas Prasloski, Alexander Rauscher, Alex L. MacKay, Madeline Hodgson, Irene M. Vavasour, Cornelia Laule, and Burkhard Mädler. Rapid whole cerebrum myelin water imaging using a 3D GRASE sequence. *NeuroImage*, 63(1):533–539, 2012. doi: 10.1016/j.neuroimage.2012.06.064.

[8] R. Marc Lebel and Alan H. Wilman. Transverse relaxometry with stimulated echo compensation. *Magnetic Resonance in Medicine*, 64(4):1005–1014, 2010. doi: 10.1002/mrm.22487.

[9] Kevin D. Harkins, Adrienne N. Dula, and Mark D. Does. Effect of intercompartmental water exchange on the apparent myelin water fraction in multiexponential T_2 measurements. *Magnetic Resonance in Medicine*, 67(3):793–800, 2012. doi: 10.1002/mrm.23053.

[10] Adrienne N. Dula, Daniel F. Gochberg, Holly L. Valentine, William M. Valentine, and Mark D. Does. Multiexponential T_2, magnetization transfer, and quantitative histology in white matter tracts of rat spinal cord. *Magnetic Resonance in Medicine*, 63(4):902–909, 2010. doi: 10.1002/mrm.22267.

[11] M. Cottaar, Z. Zheng, K.L. Miller, B.C. Tendler, and S. Jbabdi. Multi-modal Monte Carlo MRI simulator of tissue microstructure. *Imaging Neuroscience*, 2026. doi: 10.1162/IMAG.a.1177.

[12] J. Veraart, D.S. Novikov, and E. Fieremans. TE dependent diffusion imaging (TEdDI) distinguishes between compartmental T_2 relaxation times. *NeuroImage*, 182:360–369, 2018. doi: 10.1016/j.neuroimage.2017.09.030.

[13] Christoph Birkl, Jonathan Doucette, Mike Fan, Enedino Hernández-Torres, and Alexander Rauscher. Myelin water imaging depends on white matter fiber orientation in the human brain. *Magnetic Resonance in Medicine*, 85(4):2221–2231, 2021. doi: 10.1002/mrm.28543.

[14] Dmitry S. Novikov and Valerij G. Kiselev. Effective medium theory of a diffusion-weighted signal. *NMR in Biomedicine*, 23(7):682–697, 2010. doi: 10.1002/nbm.1584.

[15] Partha P. Mitra, Pabitra N. Sen, Lawrence M. Schwartz, and Pierre Le Doussal. Diffusion propagator as a probe of the structure of porous media. *Physical Review Letters*, 68(24): 3555–3558, 1992. doi: 10.1103/PhysRevLett.68.3555.

[16] J.-D. Tournier, F. Calamante, and A. Connelly. Robust determination of the fibre orientation distribution in diffusion MRI: non-negativity constrained super-resolved spherical deconvolution. *NeuroImage*, 35(4):1459–1472, 2007. doi: 10.1016/j.neuroimage.2007.02.016.

[17] F. Aboitiz, A.B. Scheibel, R.S. Fisher, and E. Zaidel. Fiber composition of the human corpus callosum. *Brain Research*, 598:143–153, 1992. doi: 10.1016/0006-8993(92)91046-W.

[18] Y. Assaf, T. Blumenfeld-Katzir, Y. Yovel, and P.J. Basser. AxCaliber: a method for measuring axon diameter distribution from diffusion MRI. *Magnetic Resonance in Medicine*, 59(6):1347–1354, 2008. doi: 10.1002/mrm.21577.

[19] P.P. Mitra, P.N. Sen, and L.M. Schwartz. Short-time behavior of the diffusion coefficient as a geometrical probe of porous media. *Physical Review B*, 47(14):8565–8574, 1993. doi: 10.1103/PhysRevB.47.8565.

[20] Aldo Zaimi, Maxime Wabartha, Victor Herman, Pierre-Louis Antonsanti, Christian S. Perone, and Julien Cohen-Adad. AxonDeepSeg: automatic axon and myelin segmentation from microscopy data using convolutional neural networks. *Scientific Reports*, 8:3816, 2018. doi: 10.1038/s41598-018-22181-4.

[21] Kathryn L. West, Nathaniel D. Kelm, Robert P. Carson, Daniel F. Gochberg, Kevin C. Ess, and Mark D. Does. Myelin volume fraction imaging with MRI. *NeuroImage*, 182: 511–521, 2018. doi: 10.1016/j.neuroimage.2016.12.067.

[22] E. Fieremans, J.H. Jensen, and J.A. Helpern. White matter characterization with diffusional kurtosis imaging. *NeuroImage*, 58:177–188, 2011. doi: 10.1016/j.neuroimage.2011.06.006.

[23] M. Barakovic, M. Pizzolato, C.M.W. Tax, U. Rudrapatna, S. Magon, T.B. Dyrby, C. Granziera, J.-P. Thiran, D.K. Jones, and E.J. Canales-Rodríguez. Estimating axon radius using diffusion-relaxation MRI: calibrating a surface-based relaxation model with histology. *Frontiers in Neuroscience*, 17:1209521, 2023. doi: 10.3389/fnins.2023.1209521.

[24] Nikola Stikov, Jennifer S.W. Campbell, Thomas Stroh, et al. In vivo histology of the myelin g-ratio with magnetic resonance imaging. *NeuroImage*, 118:397–405, 2015. doi: 10.1016/j.neuroimage.2015.05.023.

[25] H. Zhang, T. Schneider, C.A. Wheeler-Kingshott, and D.C. Alexander. NODDI: Practical in vivo neurite orientation dispersion and density imaging of the human brain. *NeuroImage*, 61:1000–1016, 2012. doi: 10.1016/j.neuroimage.2012.03.072.

[26] M. Tariq, T. Schneider, D.C. Alexander, C.A.G. Wheeler-Kingshott, and H. Zhang.

Bingham-NODDI: Mapping anisotropic orientation dispersion of neurites using diffusion MRI. *NeuroImage*, 133:207–223, 2016. doi: 10.1016/j.neuroimage.2016.01.046.

[27] E. Kaden, N.D. Kelm, R.P. Carson, M.D. Does, and D.C. Alexander. Multi-compartment microscopic diffusion imaging. *NeuroImage*, 139:346–359, 2016. doi: 10.1016/j.neuroimage.2016.06.002.

[28] Jonathan Doucette, Christian Kames, and Alexander Rauscher. DECAES – DEcomposition and Component Analysis of Exponential Signals. *Zeitschrift für Medizinische Physik*, 30(4):271–278, 2020. doi: 10.1016/j.zemedi.2020.04.001.

[29] Cornelia Laule, Irene M. Vavasour, G. R. Wayne Moore, Joel Oger, David K. B. Li, Donald W. Paty, and Alex L. MacKay. Water content and myelin water fraction in multiple sclerosis: a T2 relaxation study. *Journal of Neurology*, 251(3):284–293, 2004. doi: 10.1007/s00415-004-0306-6.

[30] Cornelia Laule, Eric Leung, David K. B. Li, Anthony L. Traboulsee, Donald W. Paty, Alex L. MacKay, and G. R. Wayne Moore. Myelin water imaging in multiple sclerosis: quantitative correlations with histopathology. *Multiple Sclerosis Journal*, 12(6):747–753, 2006. doi: 10.1177/1352458506070928.

[31] Hagen H. Kitzler, Jason Su, Michael Zeineh, Caroline Harper-Little, Alex Leung, Marcelo Kremenchutzky, Sean C. L. Deoni, and Brian K. Rutt. Deficient MWF imaging detects demyelination in the corpus callosum of multiple sclerosis patients. *NeuroImage*, 59(3): 2670–2677, 2012. doi: 10.1016/j.neuroimage.2011.08.052.

[32] E. Fieremans, L.M. Burcaw, H.-H. Lee, G. Lemberskiy, J. Veraart, and D.S. Novikov. In vivo observation and biophysical interpretation of time-dependent diffusion in human white matter. *NeuroImage*, 129:414–427, 2016. doi: 10.1016/j.neuroimage.2016.01.018.

[33] M.G. Hall and D.C. Alexander. Convergence and parameter choice for Monte-Carlo

simulations of diffusion MRI. *IEEE Transactions on Medical Imaging*, 28:1354–1364, 2009. doi: 10.1109/TMI.2009.2015756.

[34] Tianyou Xu, Sean Foxley, Michiel Kleinnijenhuis, Way Cherng Chen, and Karla L. Miller. The effect of realistic geometries on the susceptibility-weighted MR signal in white matter. *Magnetic Resonance in Medicine*, 79(1):489–500, 2018. doi: 10.1002/mrm.26689.

[35] H. Lundell, M. Nilsson, T.B. Dyrby, G.J.M. Parker, P.L. Hubbard Cristinacce, F.-L. Zhou, D. Topgaard, and S. Lasic. Multidimensional diffusion MRI with spectrally modulated gradients reveals unprecedented microstructural detail. *Scientific Reports*, 9: 9026, 2019. doi: 10.1038/s41598-019-45235-7.

[36] C.-F. Westin, H. Knutsson, O. Pasternak, F. Szczepankiewicz, E. Özarslan, D. van Westen, and M. Nilsson. Q-space trajectory imaging for multidimensional diffusion MRI of the human brain. *NeuroImage*, 135:345–362, 2016. doi: 10.1016/j.neuroimage.2016.02.039.

[37] M.D. Does, E.C. Parsons, and J.C. Gore. Oscillating gradient measurements of water diffusion in normal and globally ischemic rat brain. *Magnetic Resonance in Medicine*, 49:206–215, 2003. doi: 10.1002/mrm.10385.

[38] M. Nilsson, D. van Westen, F. Ståhlberg, P.C. Sundgren, and J. Lätt. The role of tissue microstructure and water exchange in biophysical modelling of diffusion in white matter. *MAGMA*, 26:345–370, 2013. doi: 10.1007/s10334-013-0371-x.

[39] D.S. Grebenkov. NMR survey of reflected Brownian motion. *Reviews of Modern Physics*, 79:1077–1137, 2007. doi: 10.1103/RevModPhys.79.1077.

[40] R.H.J. Fick, D. Wassermann, and R. Deriche. The dmipy toolbox: diffusion MRI multi-compartment modeling and microstructure recovery made easy. *Frontiers in Neuroinformatics*, 13:64, 2019. doi: 10.3389/fninf.2019.00064.